

Project 1- scRNAseq

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Week 1

1.3 Add Meta-data

1. How many cells are in each sample?

BMMC-D1T1 >> 6270

BMMC-D1T2 >> 6332

CD34-D2T1 >> 2424

CD34-D3T1 >> 5752

2. How many genes are in the expression matrices?

BMMC-D1T1 >> 20287

BMMC-D1T2 >> 20287

CD34-D2T1 >> 20287

CD34-D3T1 >> 20287

3. What information is now part of the meta-data of the objects?

For each we have { orig.ident / nCount_RNA / nFeature_RNA / Donor / Replicate / Sex / percent.mt)

Week 2

4.1 Preprocessing

Before merging (filtering, normalization, feature selection , doublet removal)

After merging (batch correction)

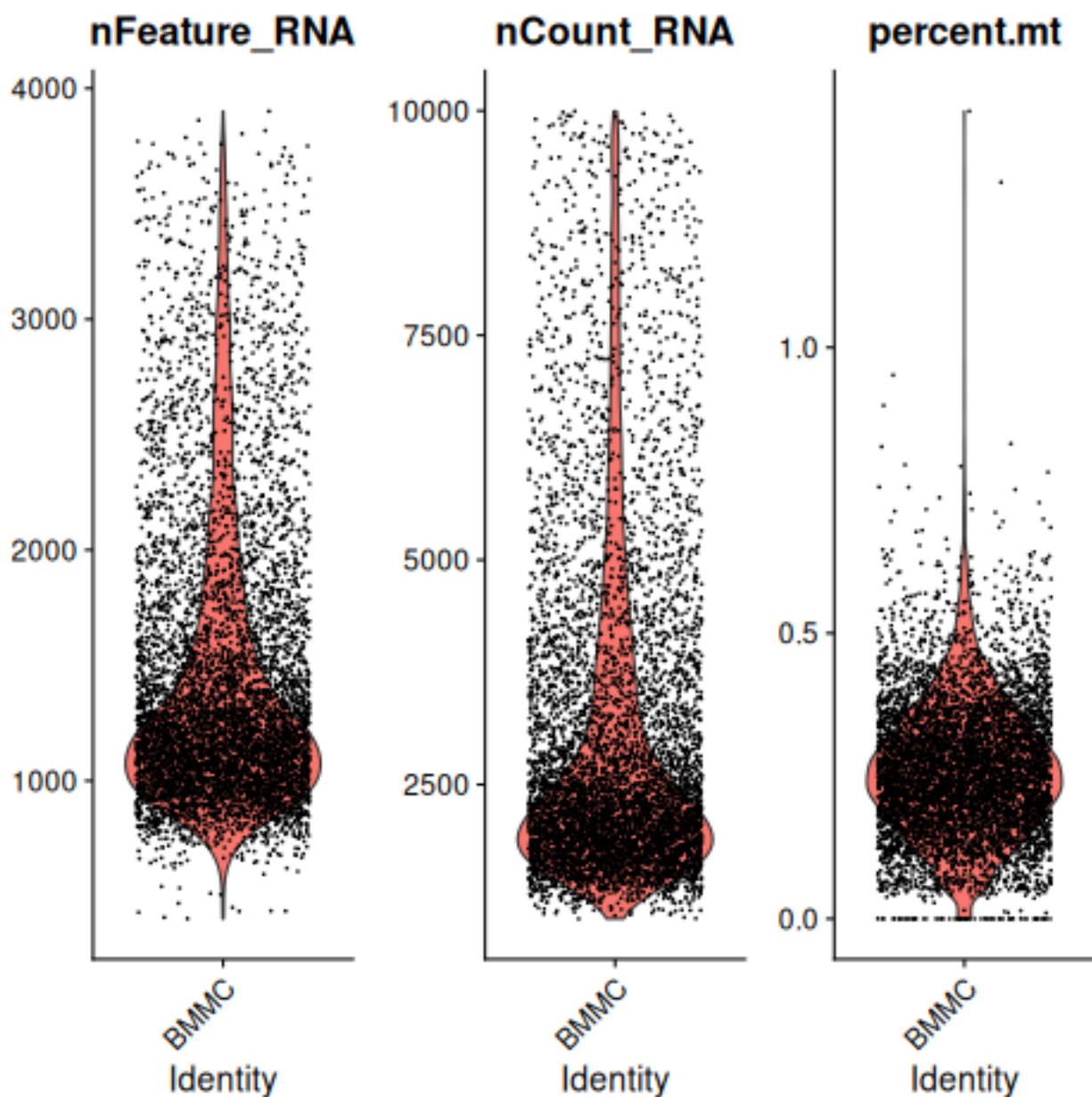
1. Filtering: The parameter used for filtering is `nfeature_RNA`, or gene count. We utilized this parameter because we wanted to count the number of unique features. Cut-off parameters are determined using the scatter and violin plots.

The frequency of values is determined by the Violin plot's regions. The violin plot shows how our characteristics and RNA counts are dispersed and where our data is concentrated.

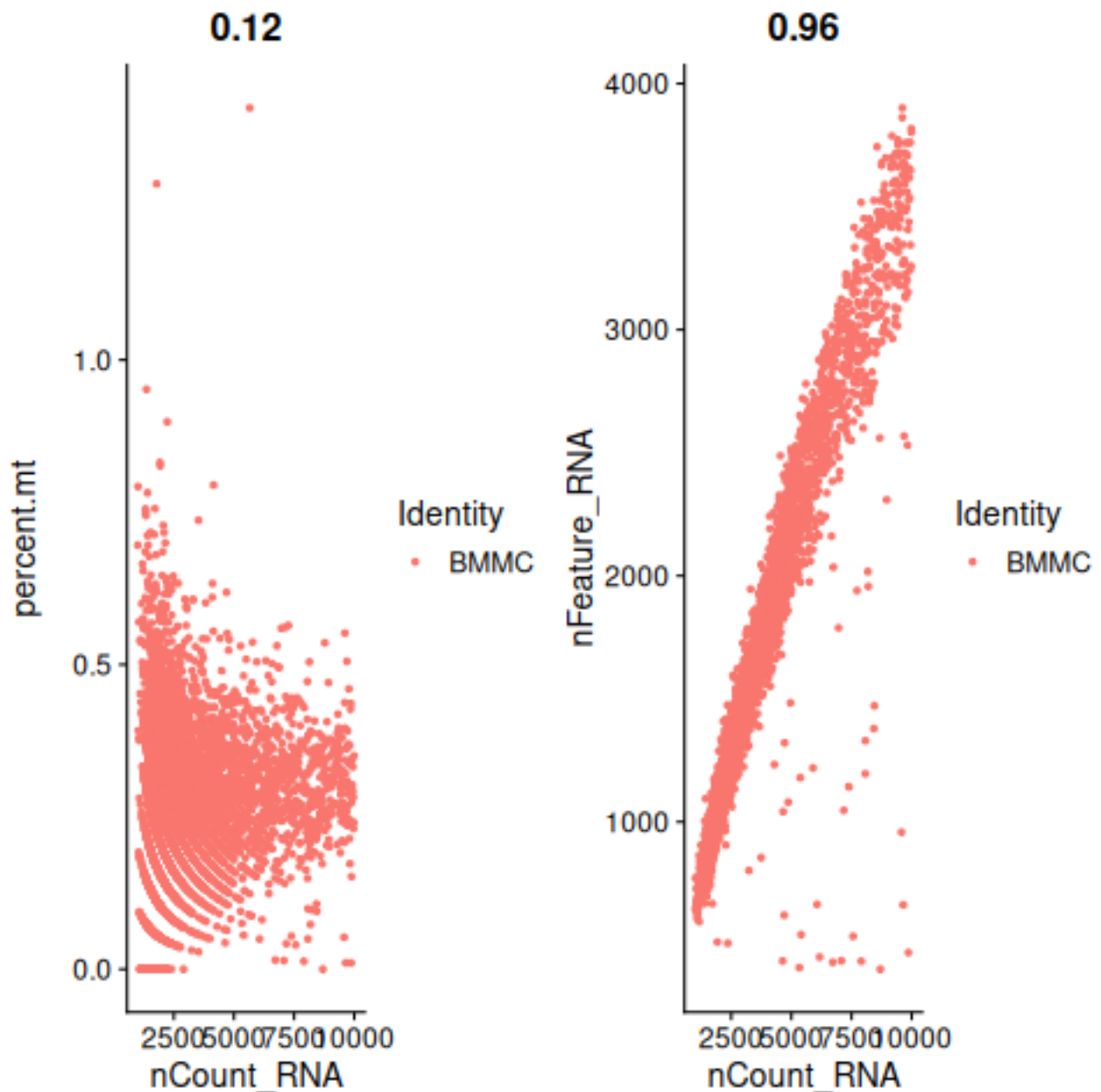
For example, in the image below, the Violin plot for `percent.mt` is a flat line because our data does not include mitochondrial genes. The distribution of gene count and RNA count are centered about 250 and 2500, respectively.

```
#bmmc_d1t1
```

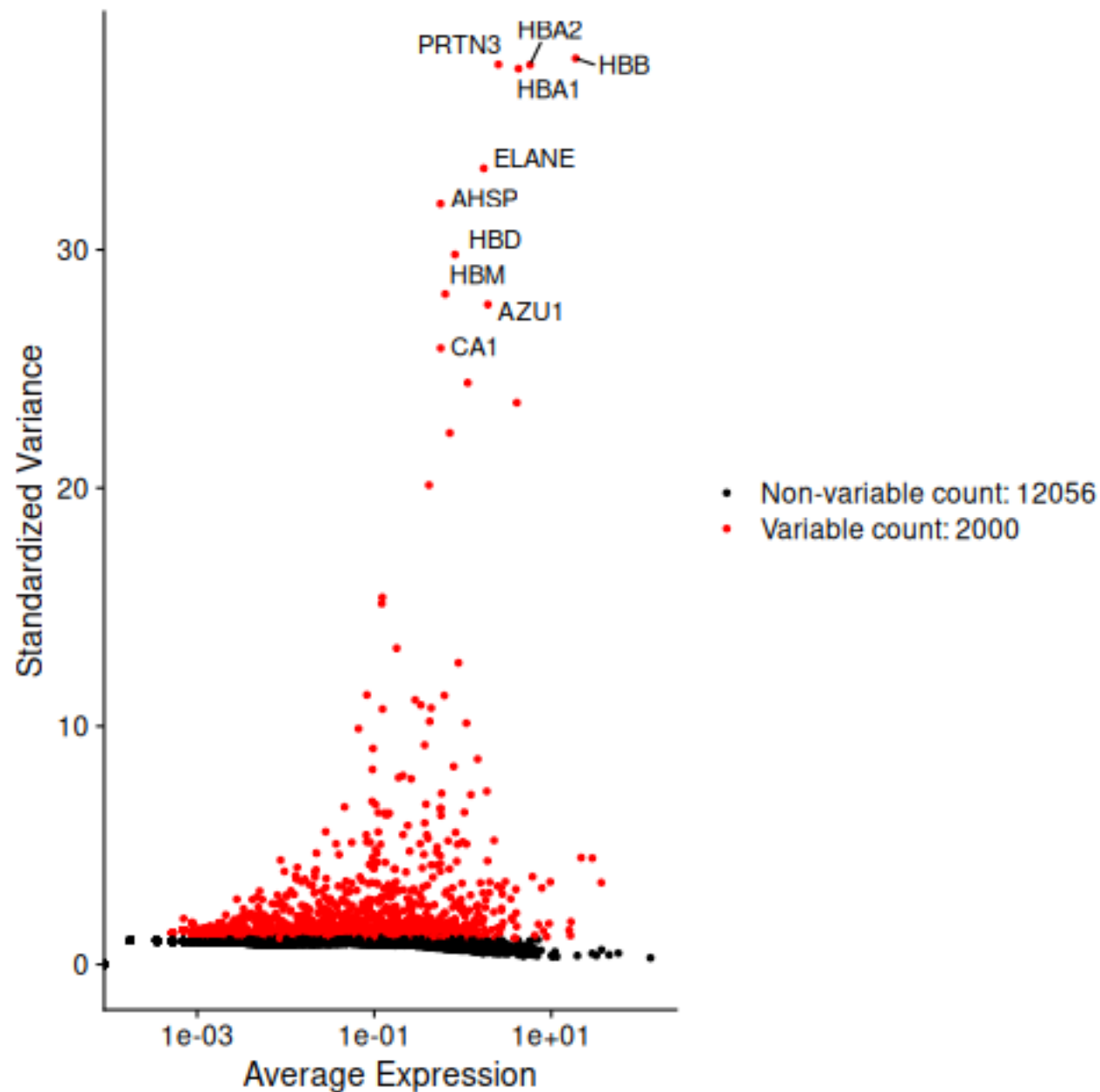
Violin plot for bmmc_d1t1



The scatter figure shows the relationship between percent.mt vs RNA count and gene count vs RNA count. We can plainly observe that there is no relationship with percent.mt because our data contains no mitochondrial genes, however there is a robust relationship with gene count.

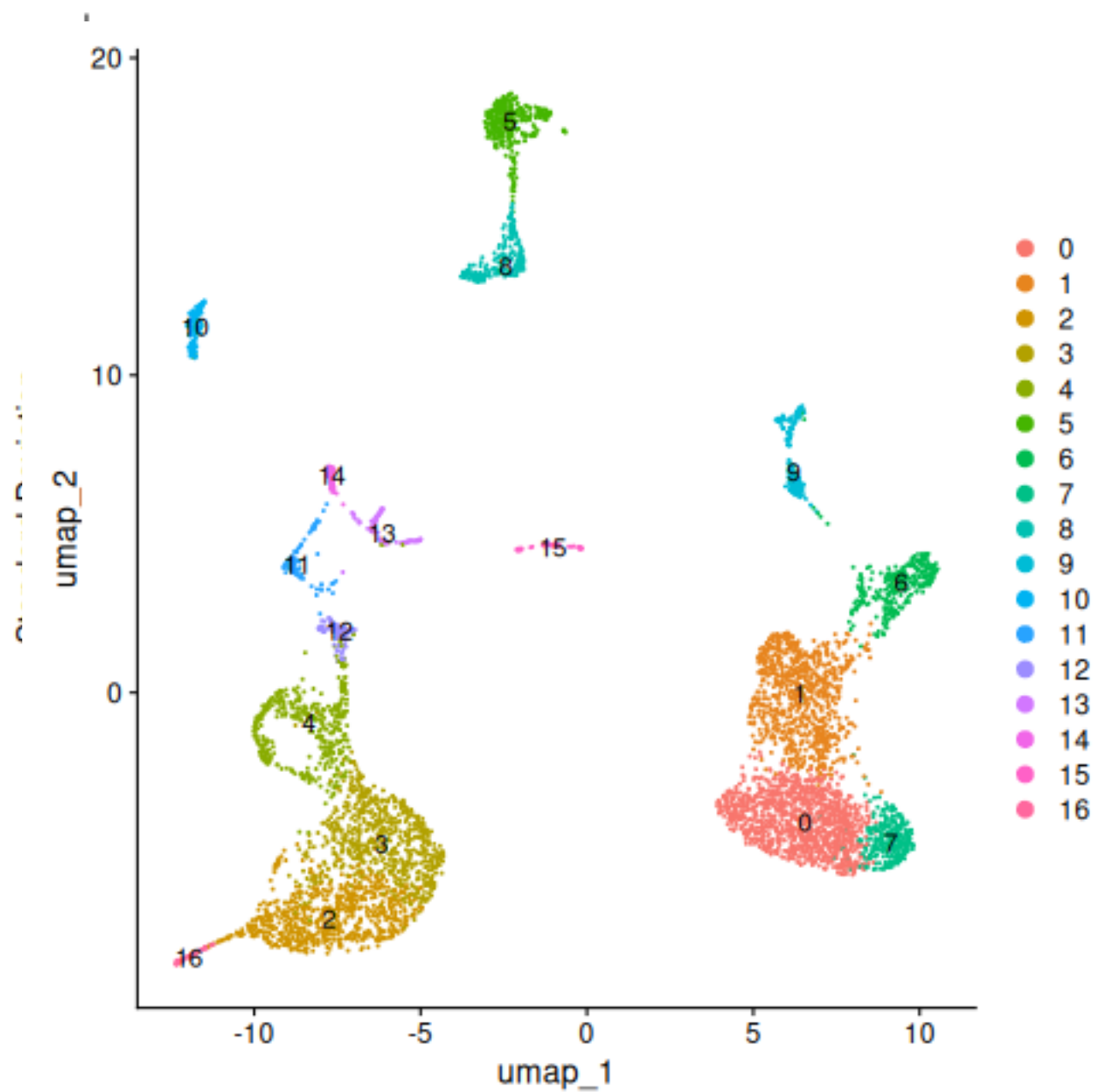


2-Normalization: After identifying interesting cells, we normalized the data. A global-scaling normalization method, LogNormalize (by default), with a scale factor of 10,000.

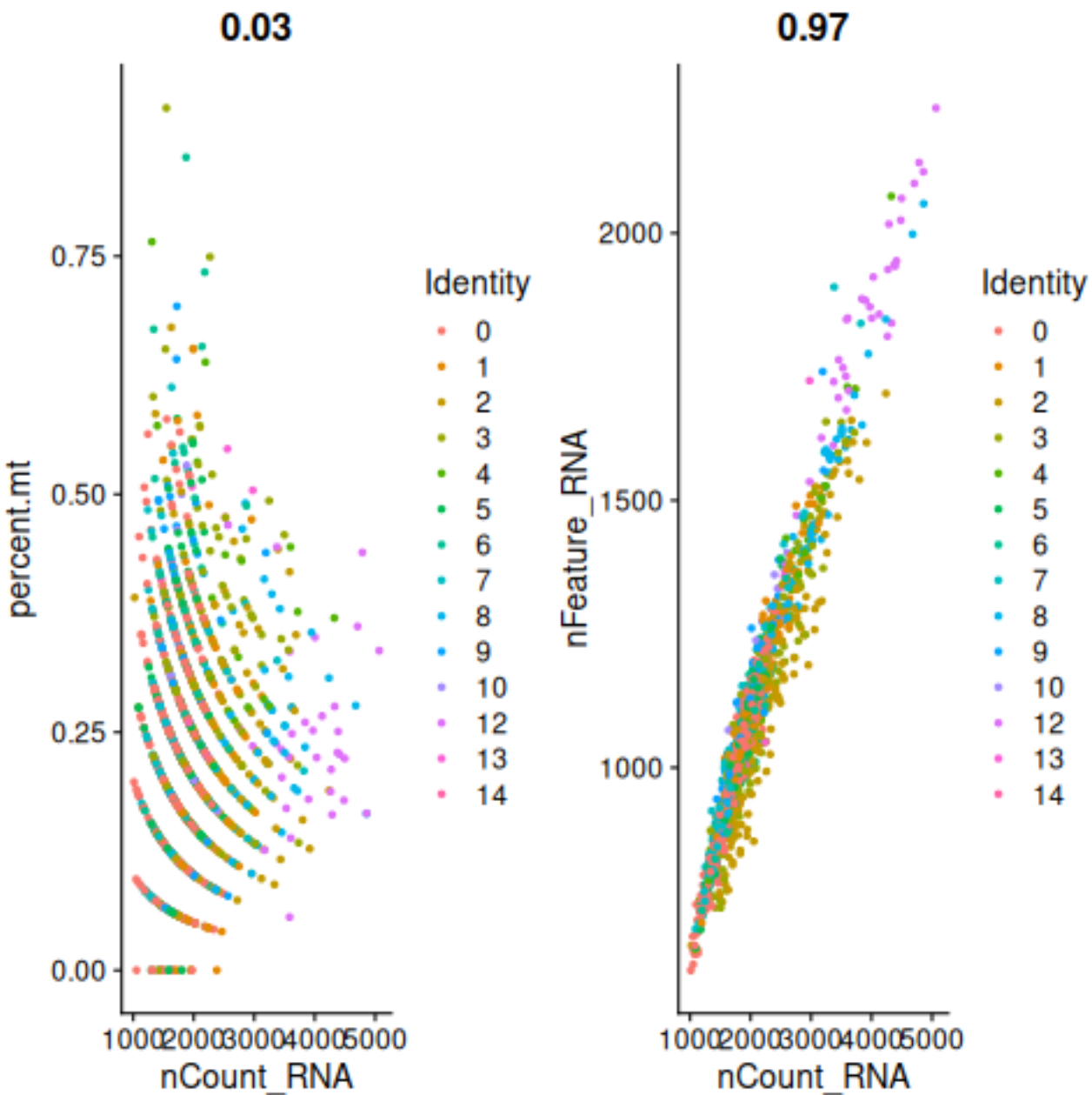


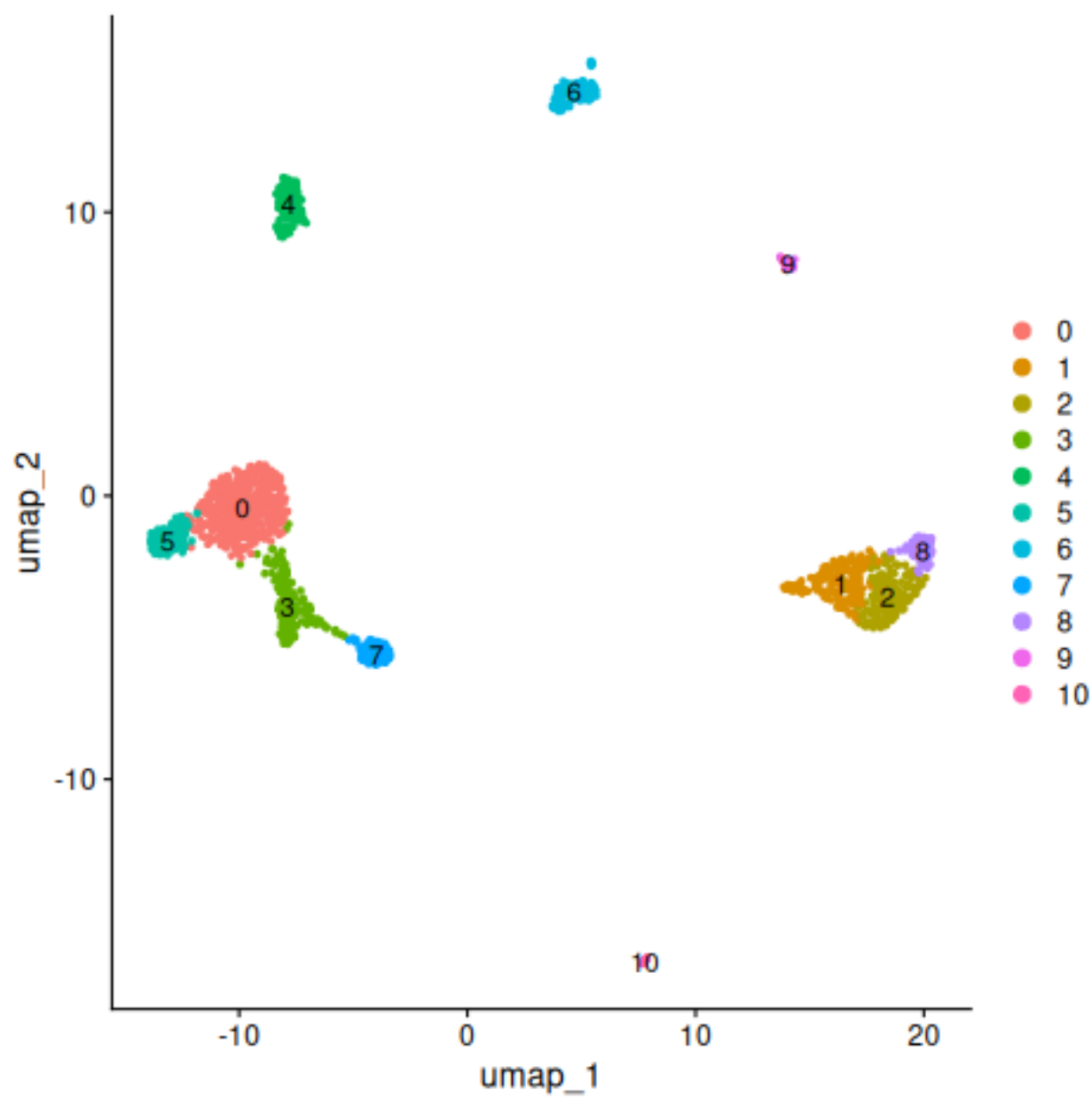
3- Feature selection: In this step, we require features that exhibit significant cell-to-cell variance. By default, we take 2000 subsets per dataset. The label genes in the figures below have the highest variation, indicating that they are highly variable genes

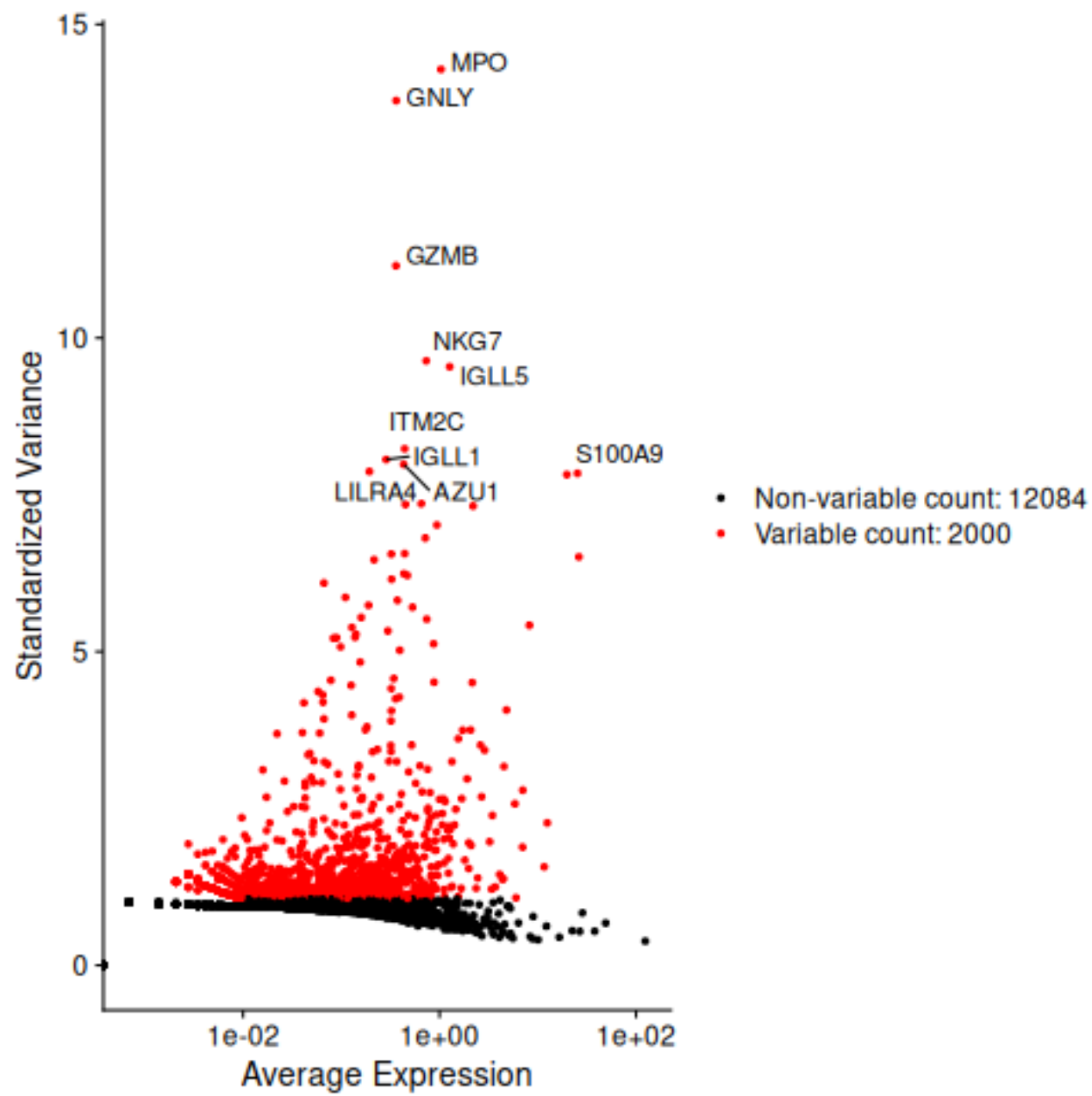
4-Doublet removal: The pK value forecasts the PC neighbor size. We calculated the best value using pN-pK parameter sweeps (paramSweep in R). After calculating the pK value, we apply mean-variance normalization. The figures below show the ideal pK value, which has the largest peak.



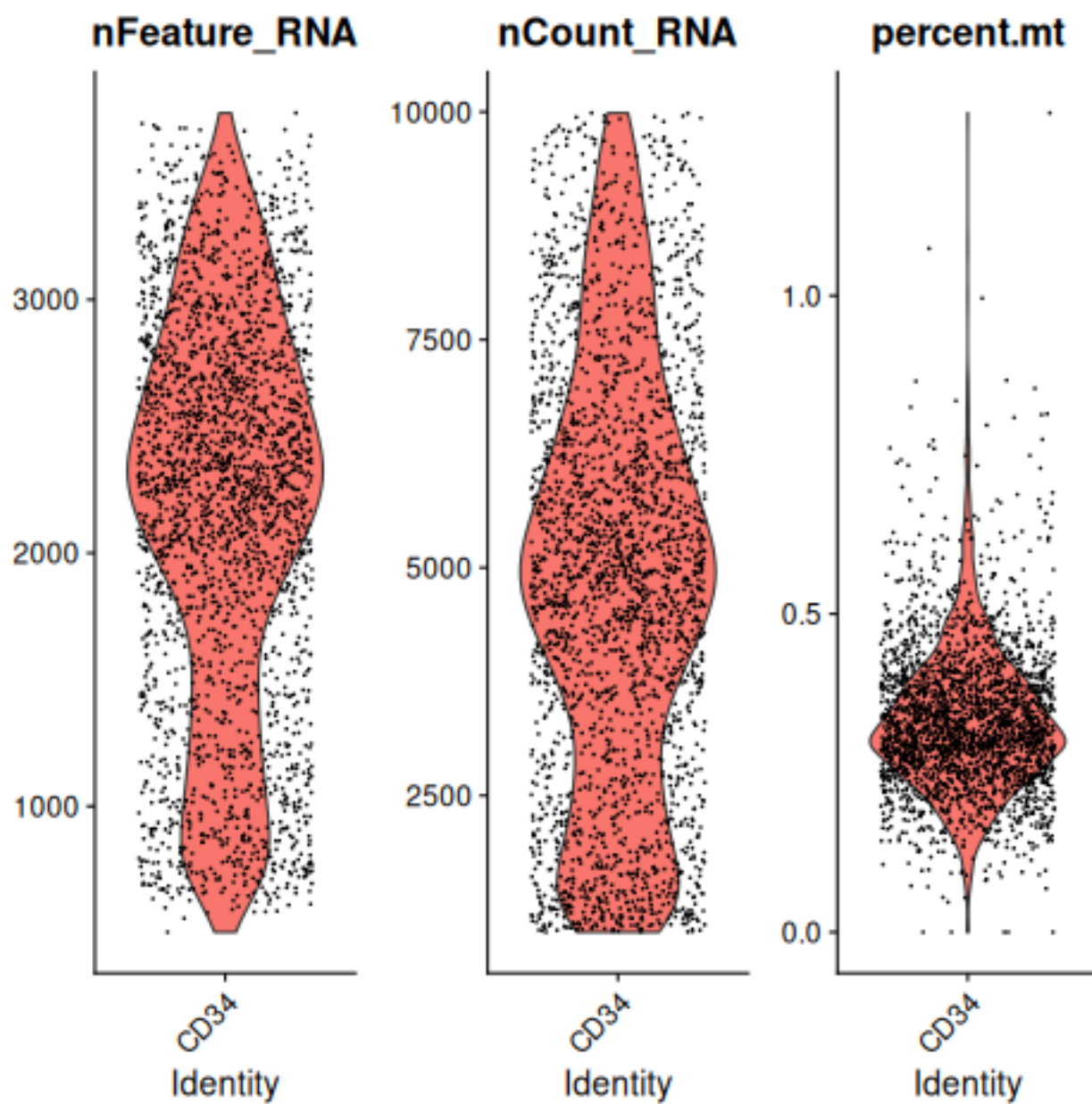
#bmmc_d1t2





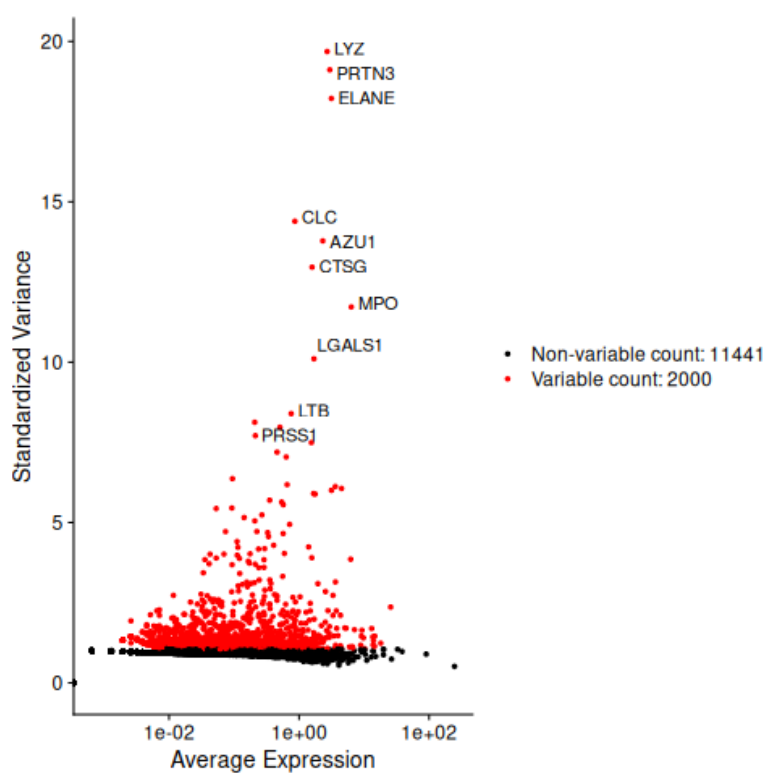
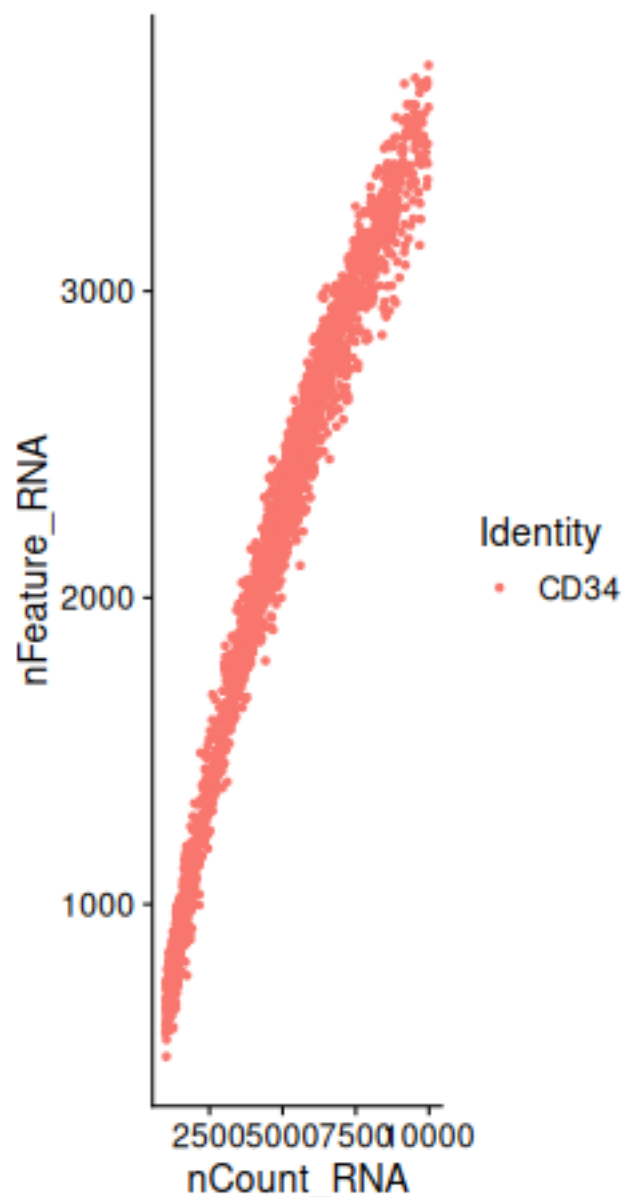
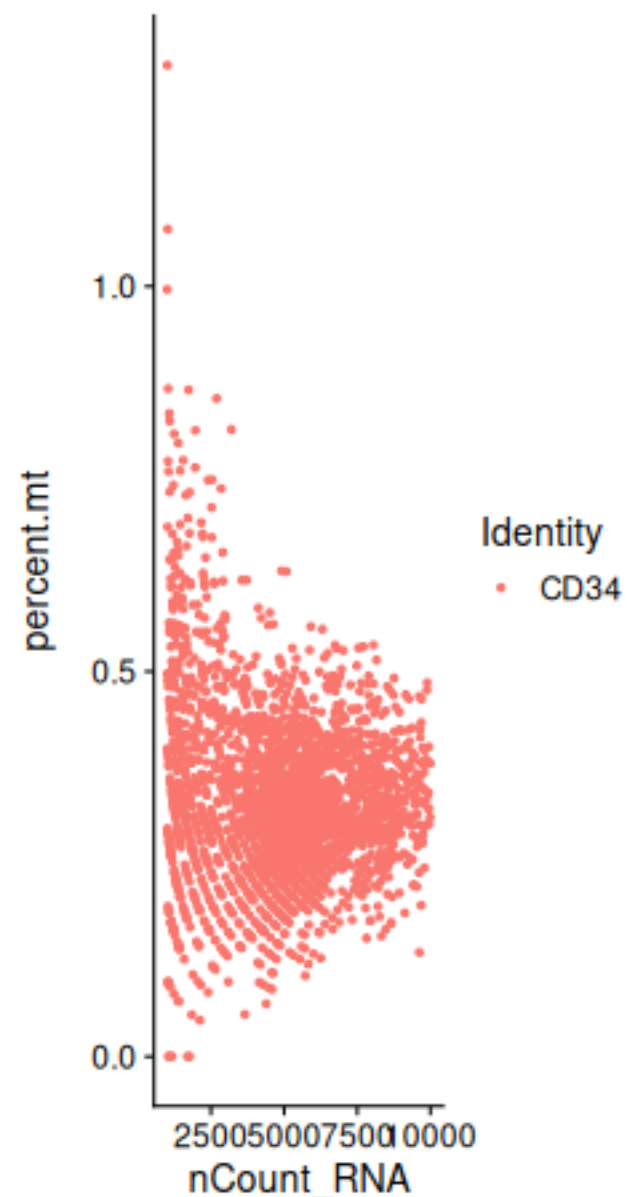


#CD34_D2T1

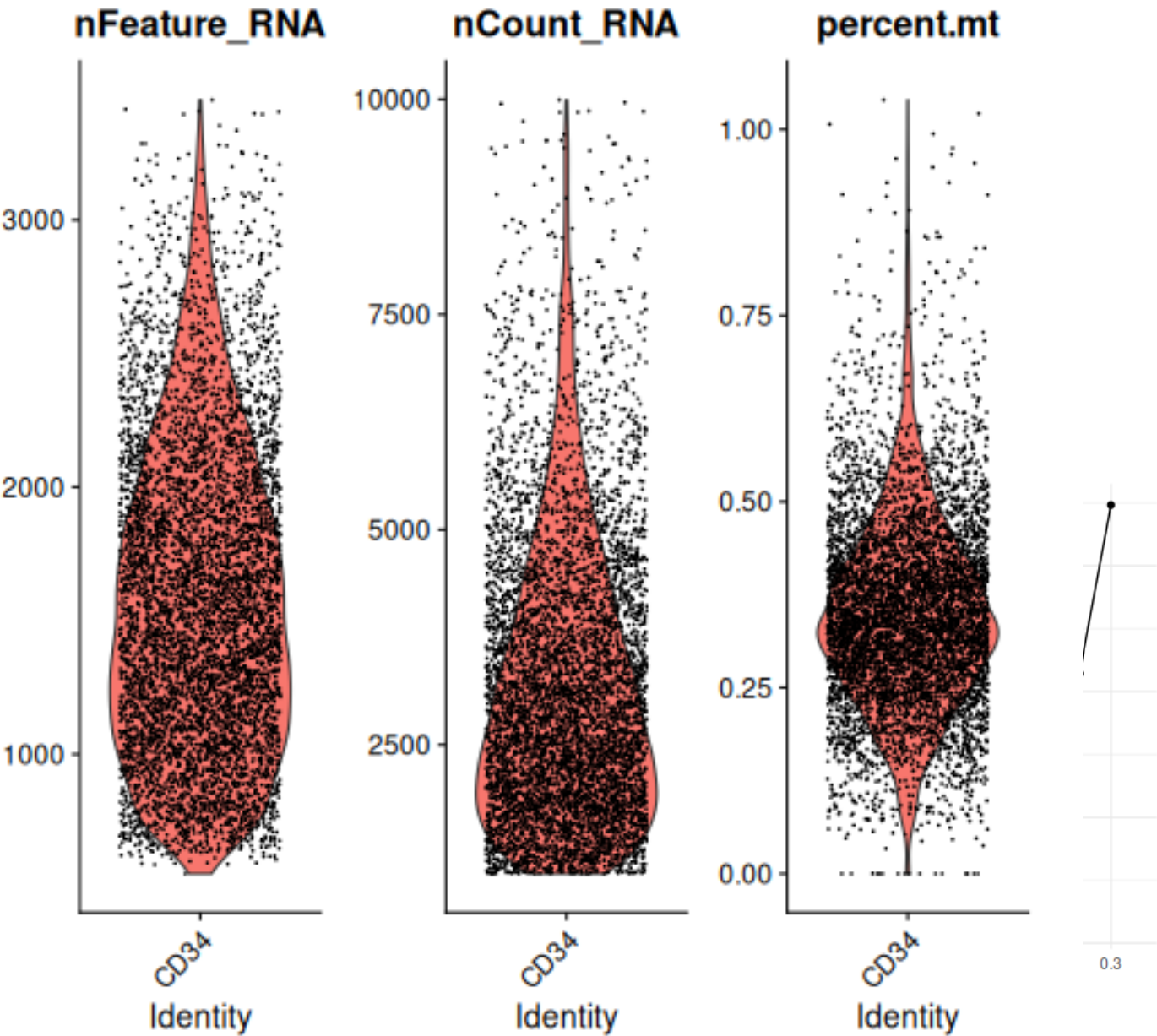


-0.12

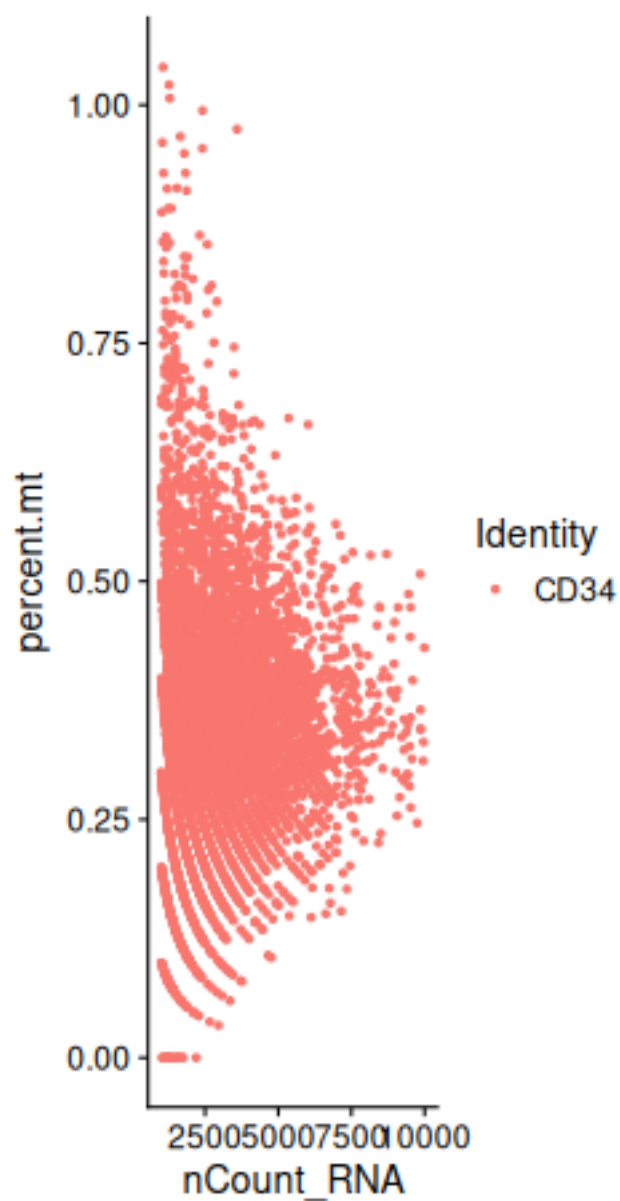
0.98



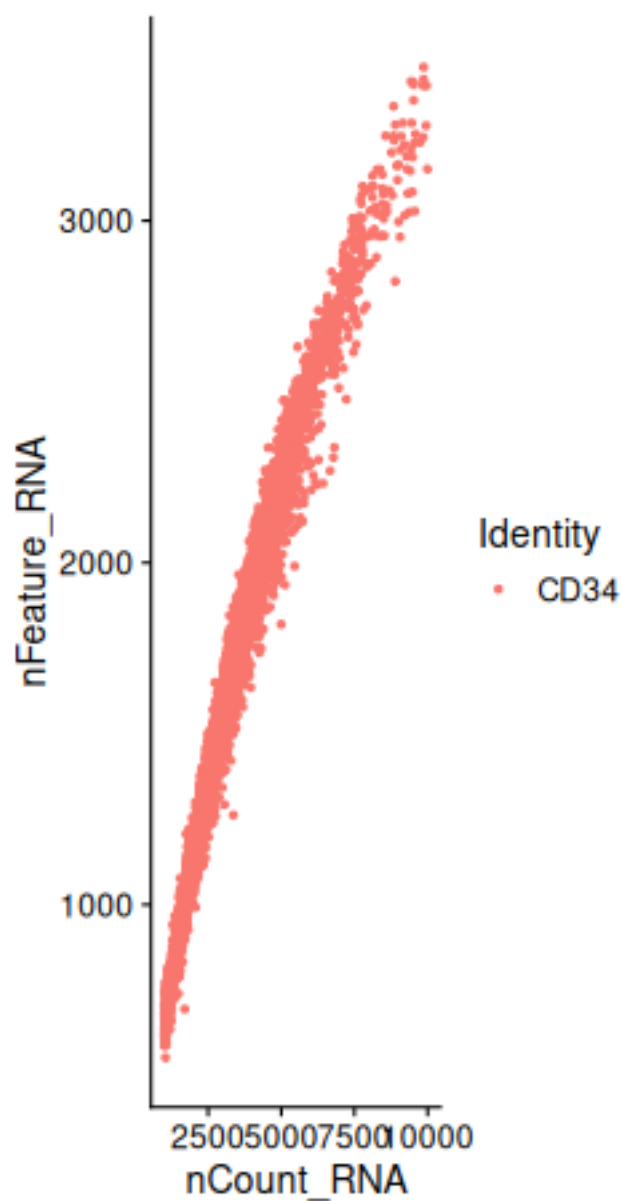
#CD34_D3T1

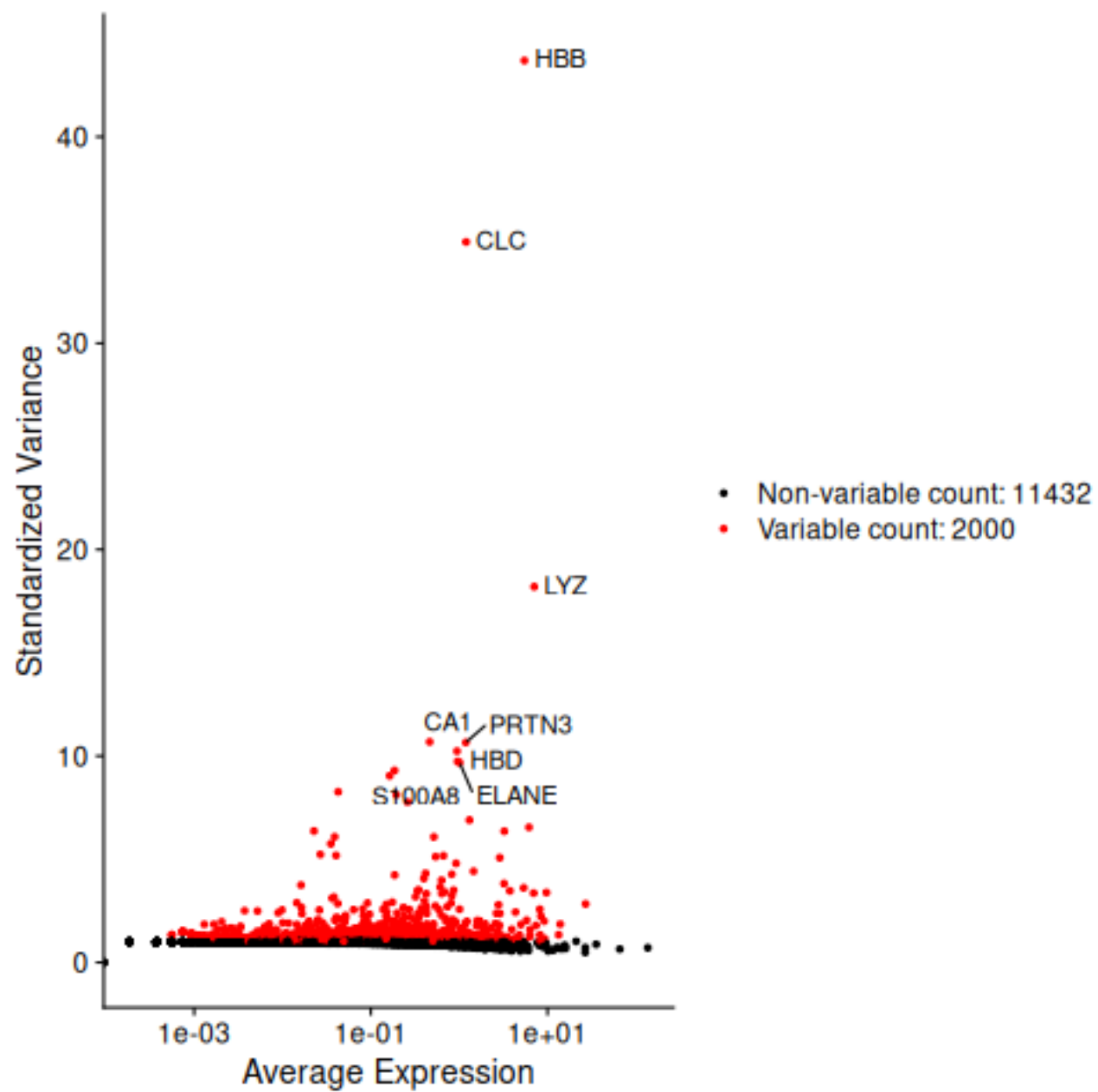


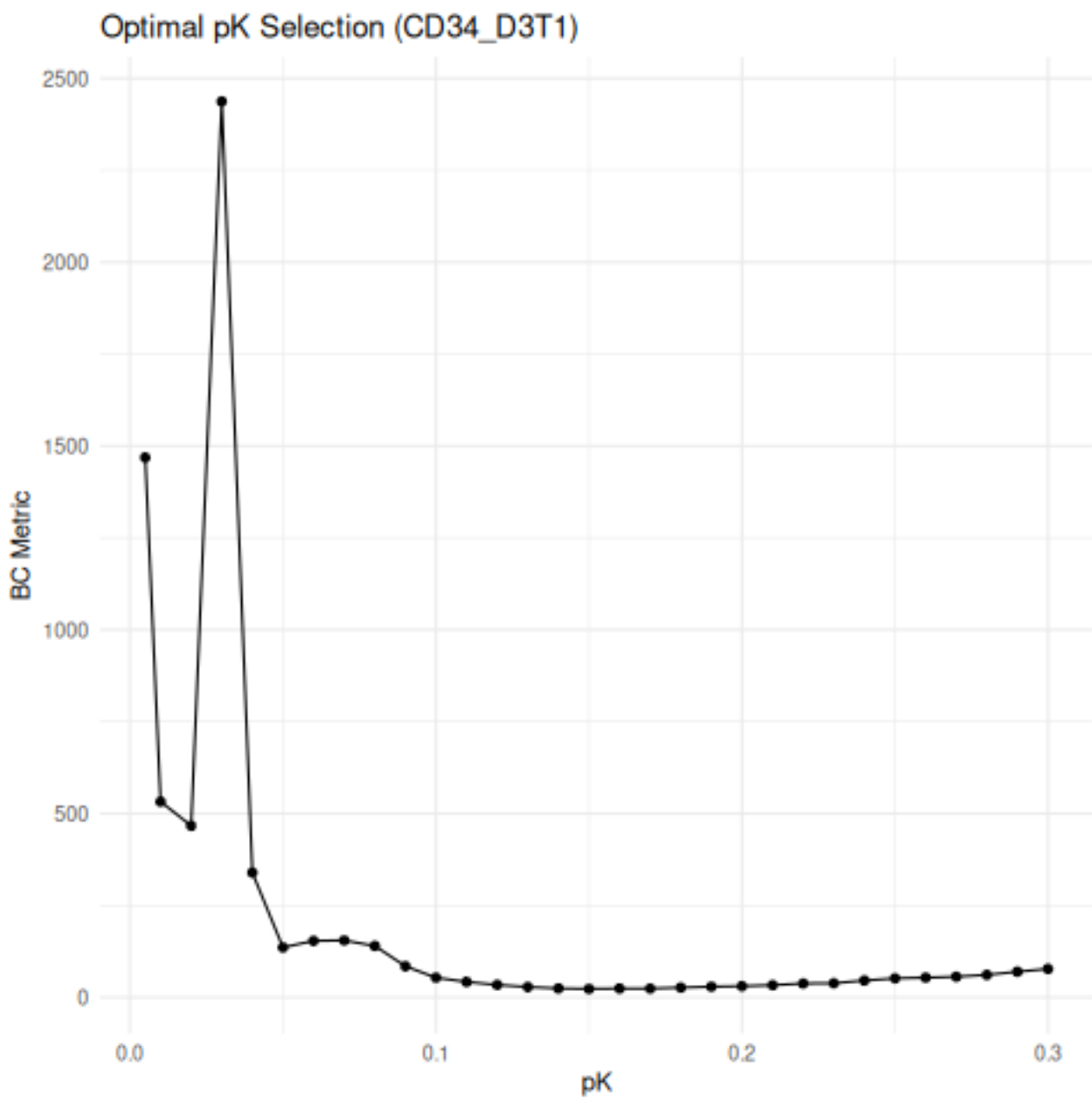
0.01

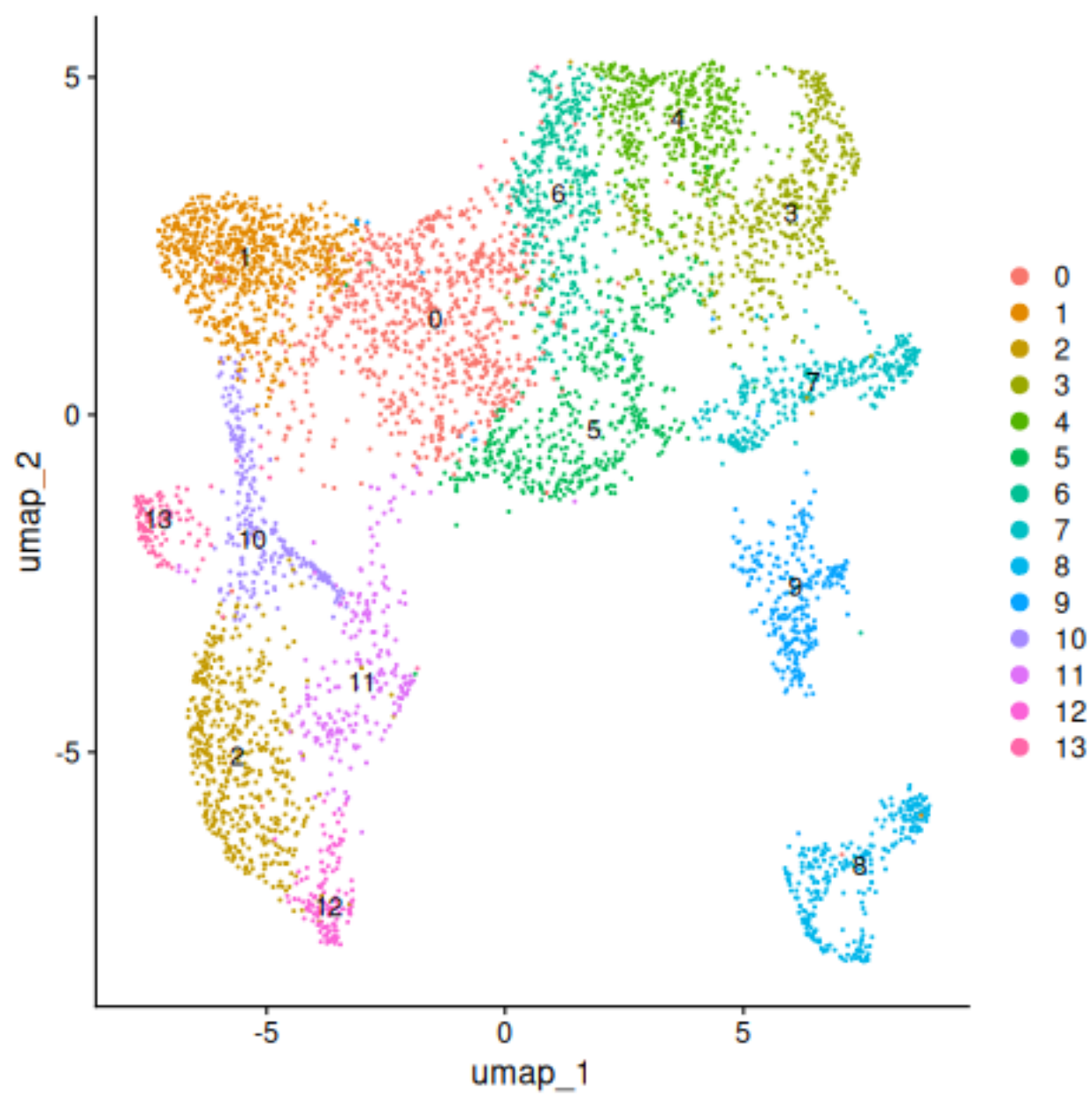


0.98

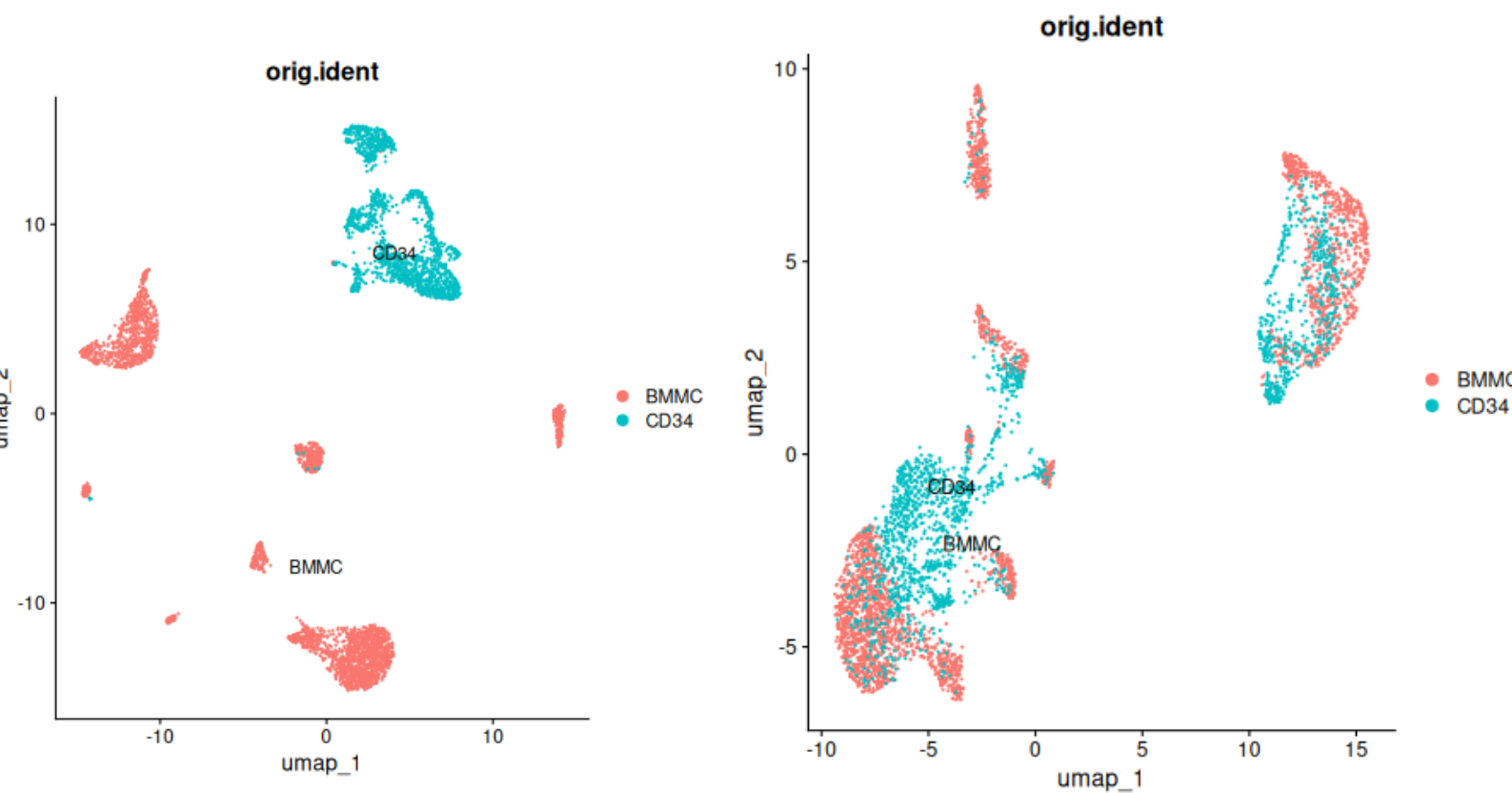




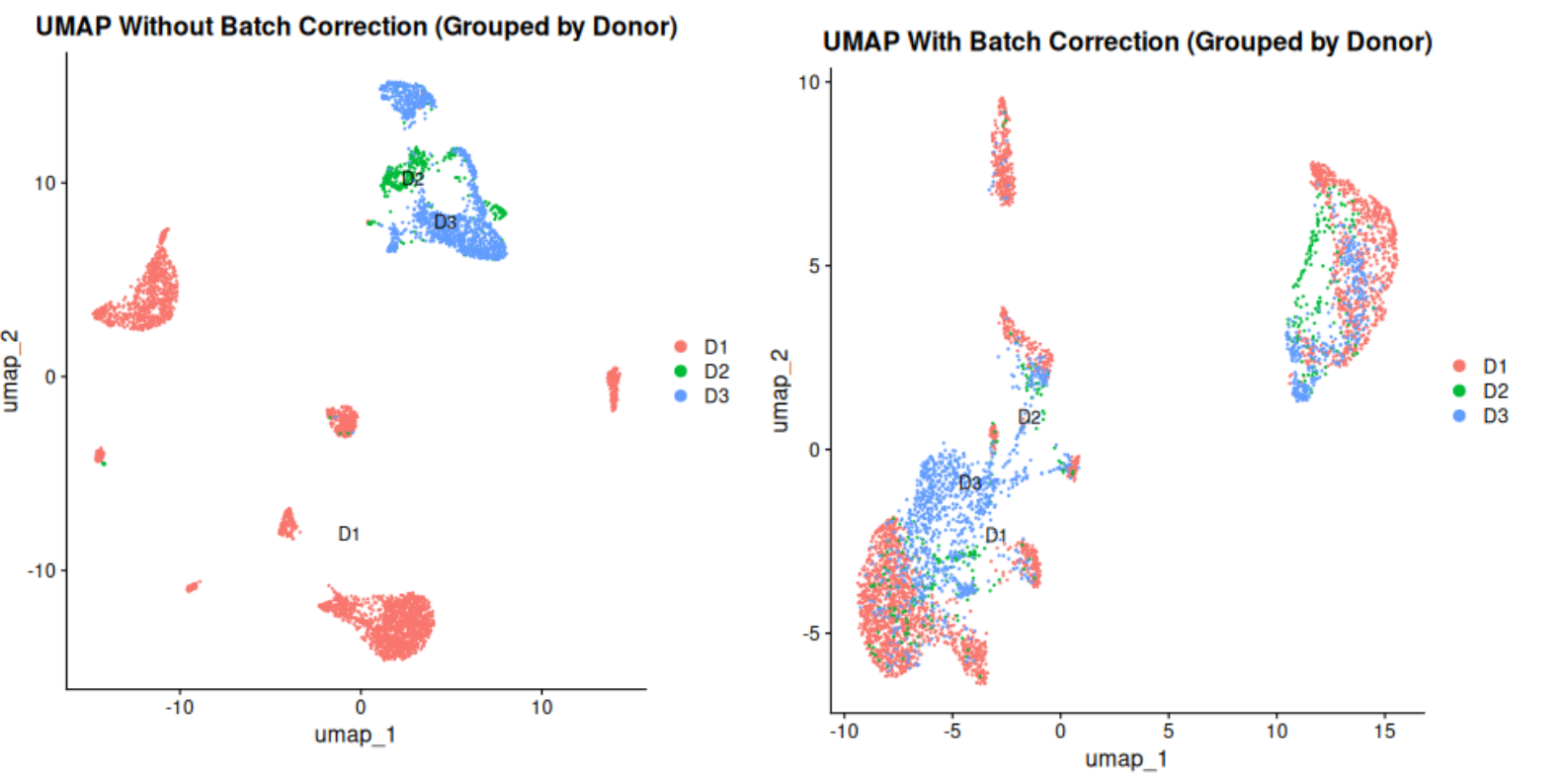




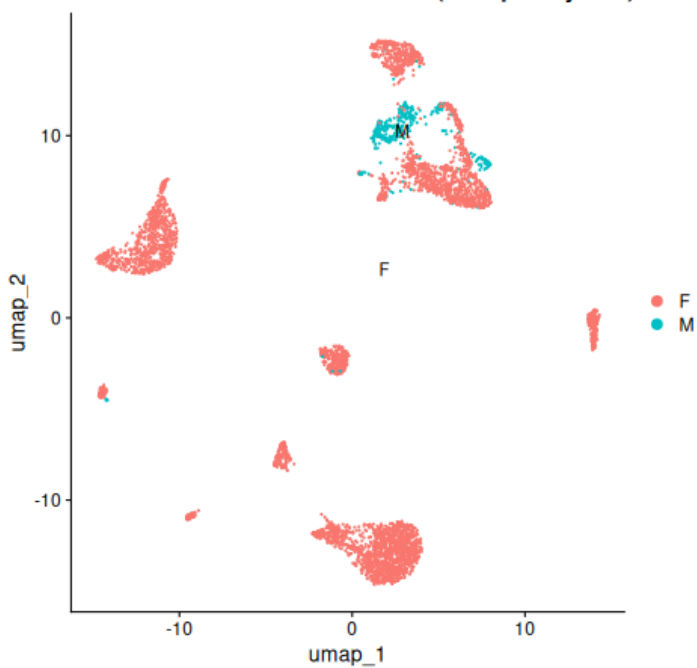
Task 2.2: Batch-Correction



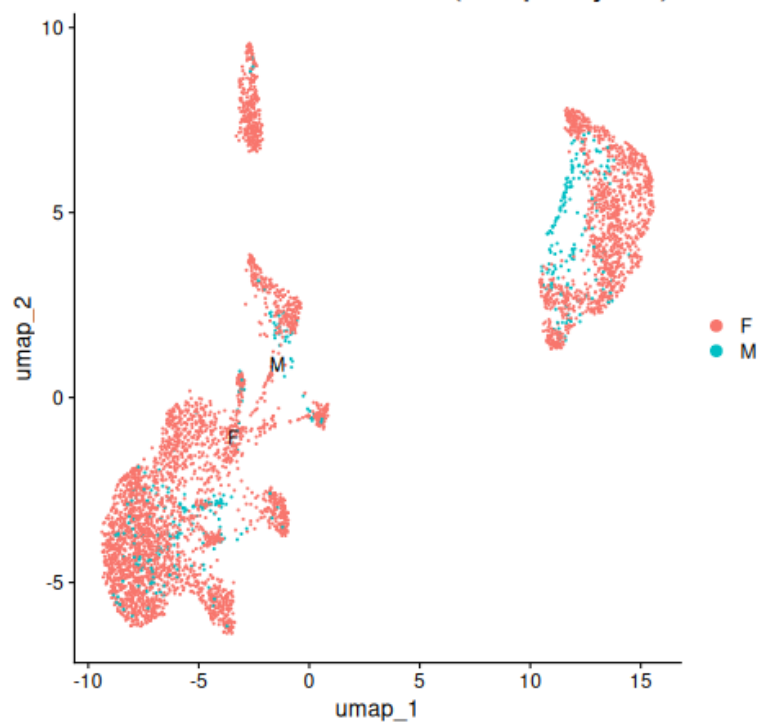
a)comparison between cell types



UMAP Without Batch Correction (Grouped by Sex)

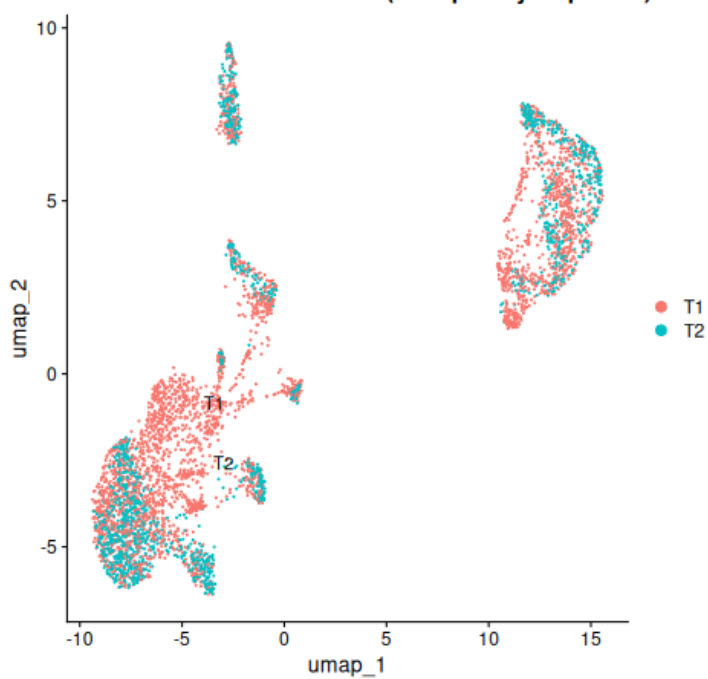


UMAP With Batch Correction (Grouped by Sex)

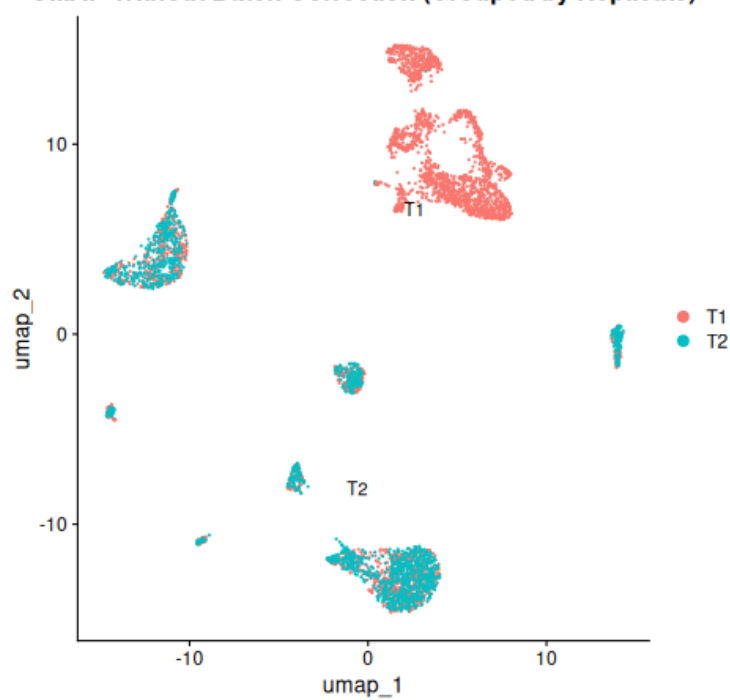


c) comparison between sex type

UMAP With Batch Correction (Grouped by Replicate)



UMAP Without Batch Correction (Grouped by Replicate)

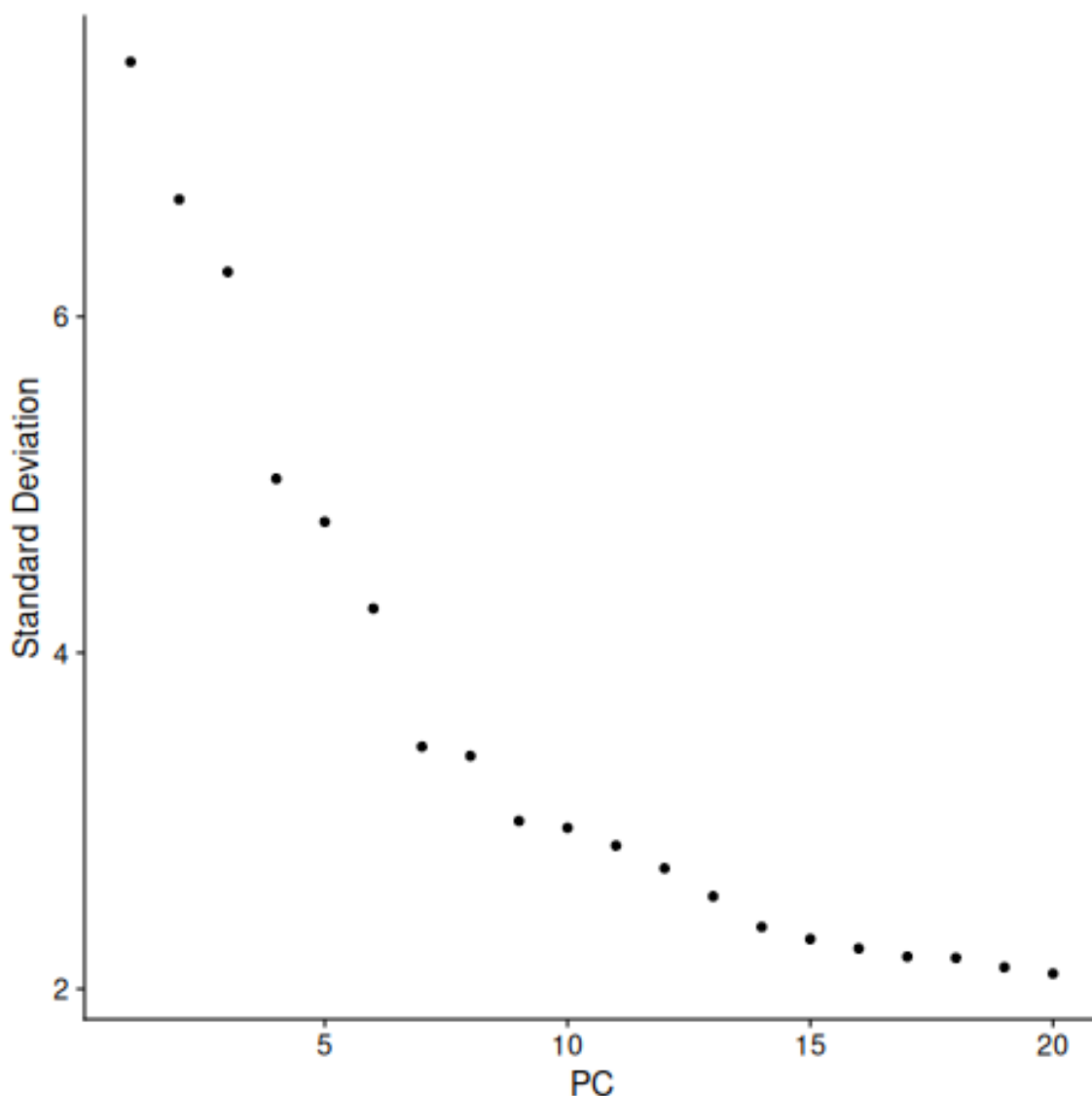


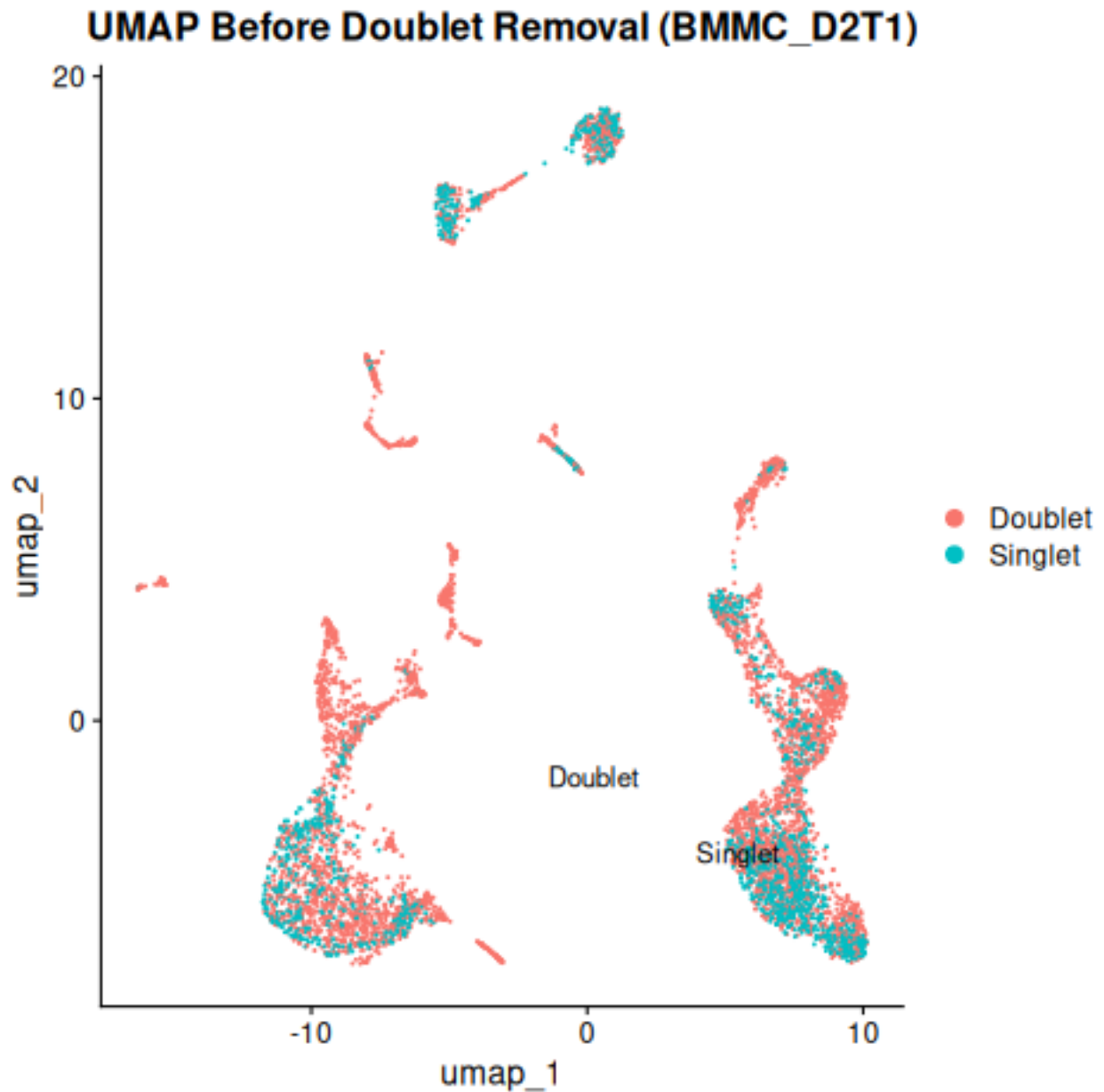
d)comparison between replicate type

The plots above show the split batches (before batch correction) and the merging of some cells (after batch correction) in distinct graphs. However, we can still see batches after batch correction, which could be owing to the fact that these cells differ from one another or a UMI error. Yes, batch correction is required for the variables orig.ident and Replicate, as evidenced by the data above, which show partial cell merging but not complete removal. Other variables do not show substantial cell merging.

Dimensionality Reduction

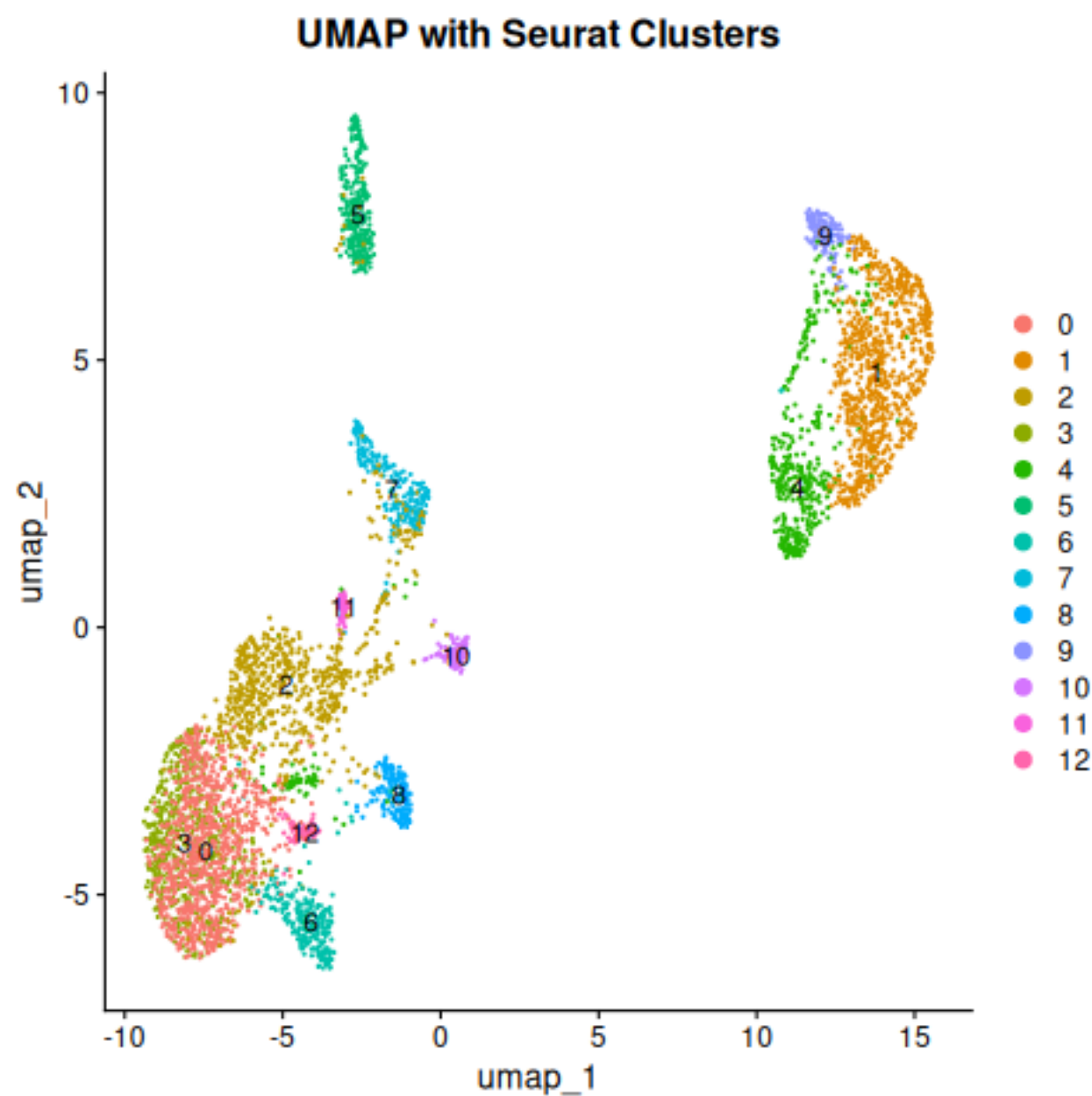
We picked the number of dimensions based on the number of principal components discovered. The figure below (elbow plot) predicts the number of PCs in our data. PCs with higher standard deviation are chosen to be dimension 1, and so on. The first ten PCs catch the real signals, which indicate an elbow around that location.





PCA with UMAP is utilized because PCA is used before UMAP for additional analysis and dimension reduction. UMAP then scales the distance between PCs, reducing the dimension to two. If only UMAP is utilized, it may be difficult to read, but it does preserve the data structure. And if simply PCA is employed, the data's significant signals may be lost, but it is highly interpretable.

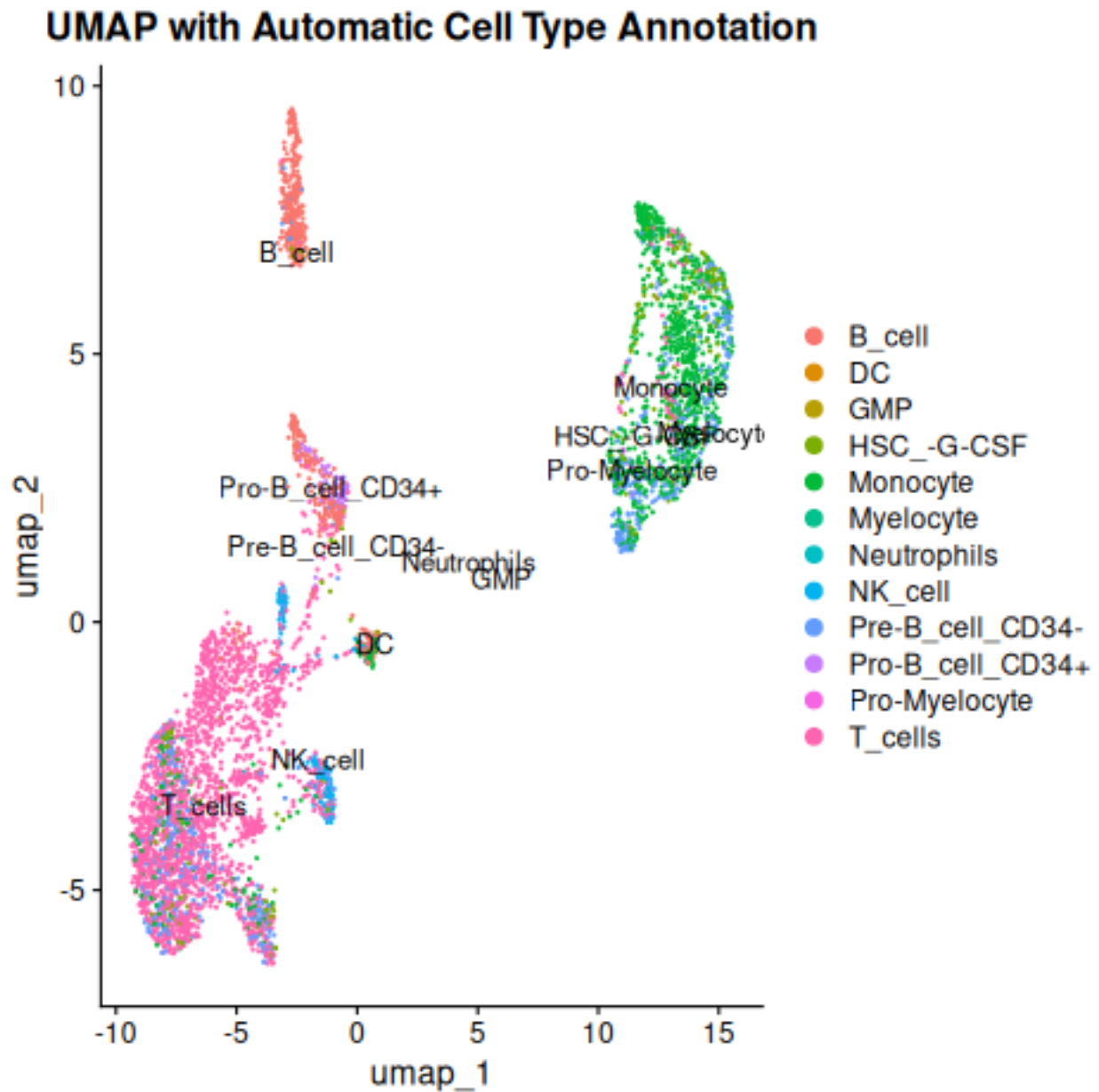
Clustering



Week 3

Cell Type Annotation

Automatic Annotation



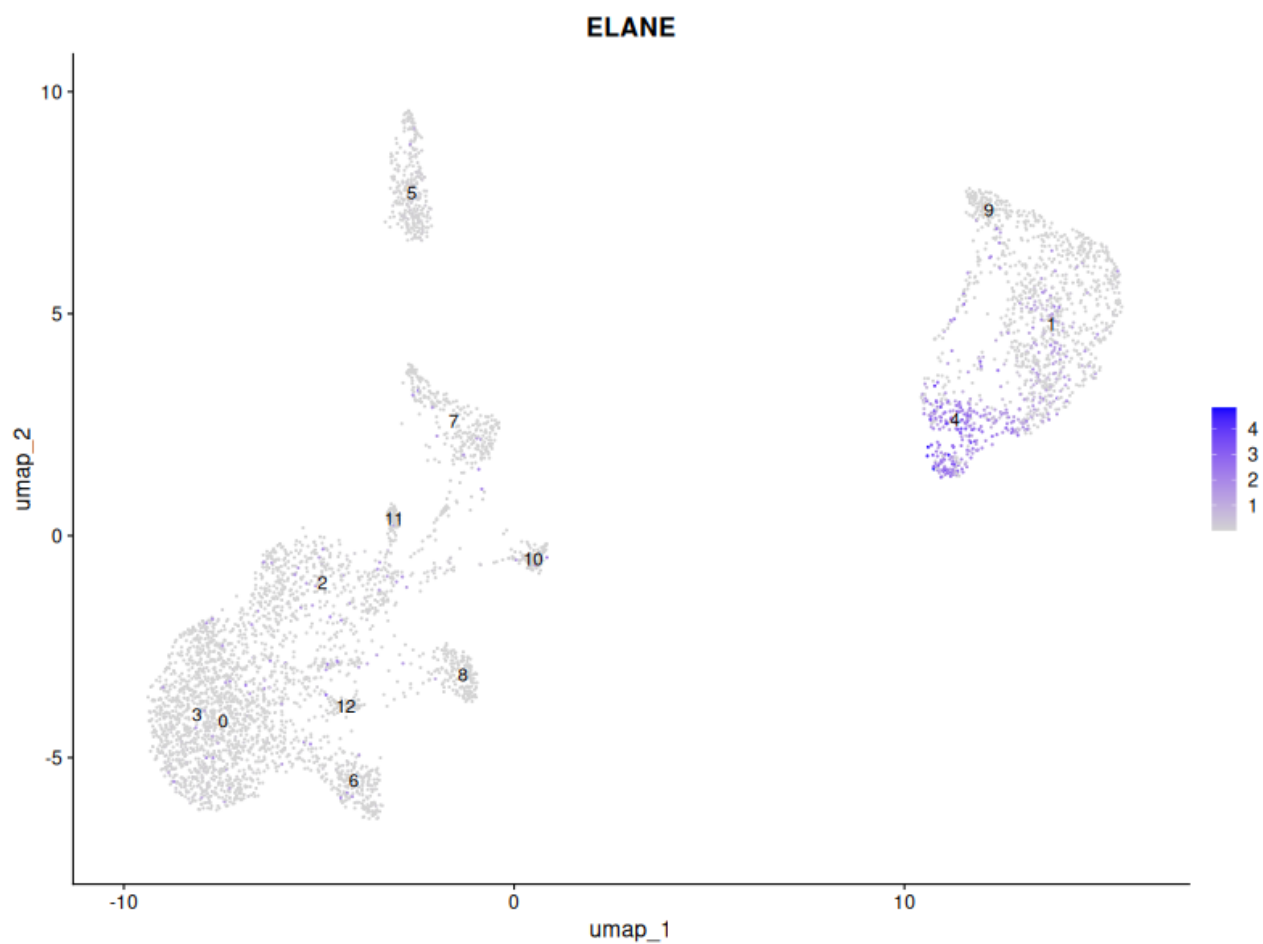
Manual Annotation

a) CD8 scatter plot

b) b) CD4 scatter plot

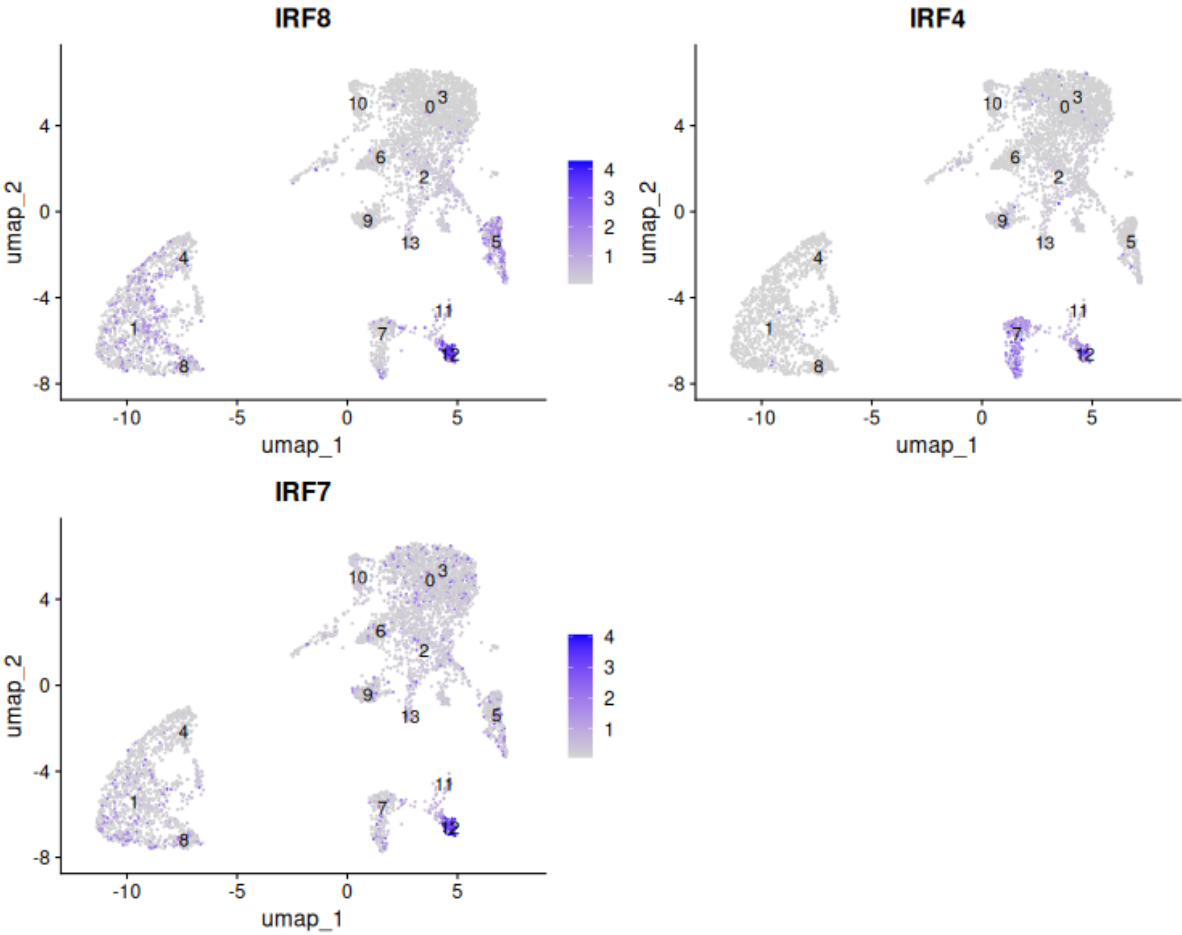
c) GMP scatter plot:

Based on GMP plot it seems that this marker is more expressed at cluster 4

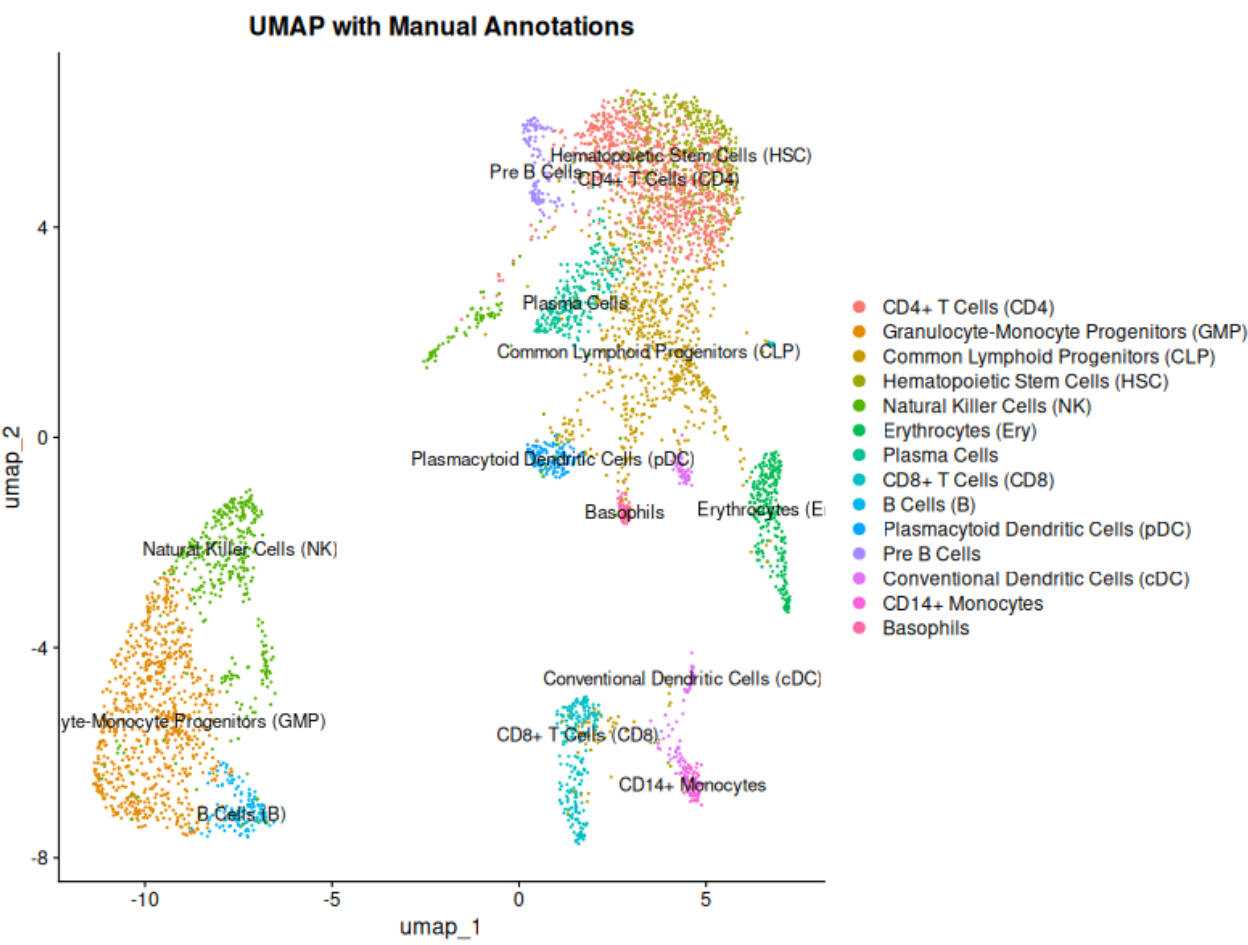


d) pDC scatter plot:

Clusters 12,7

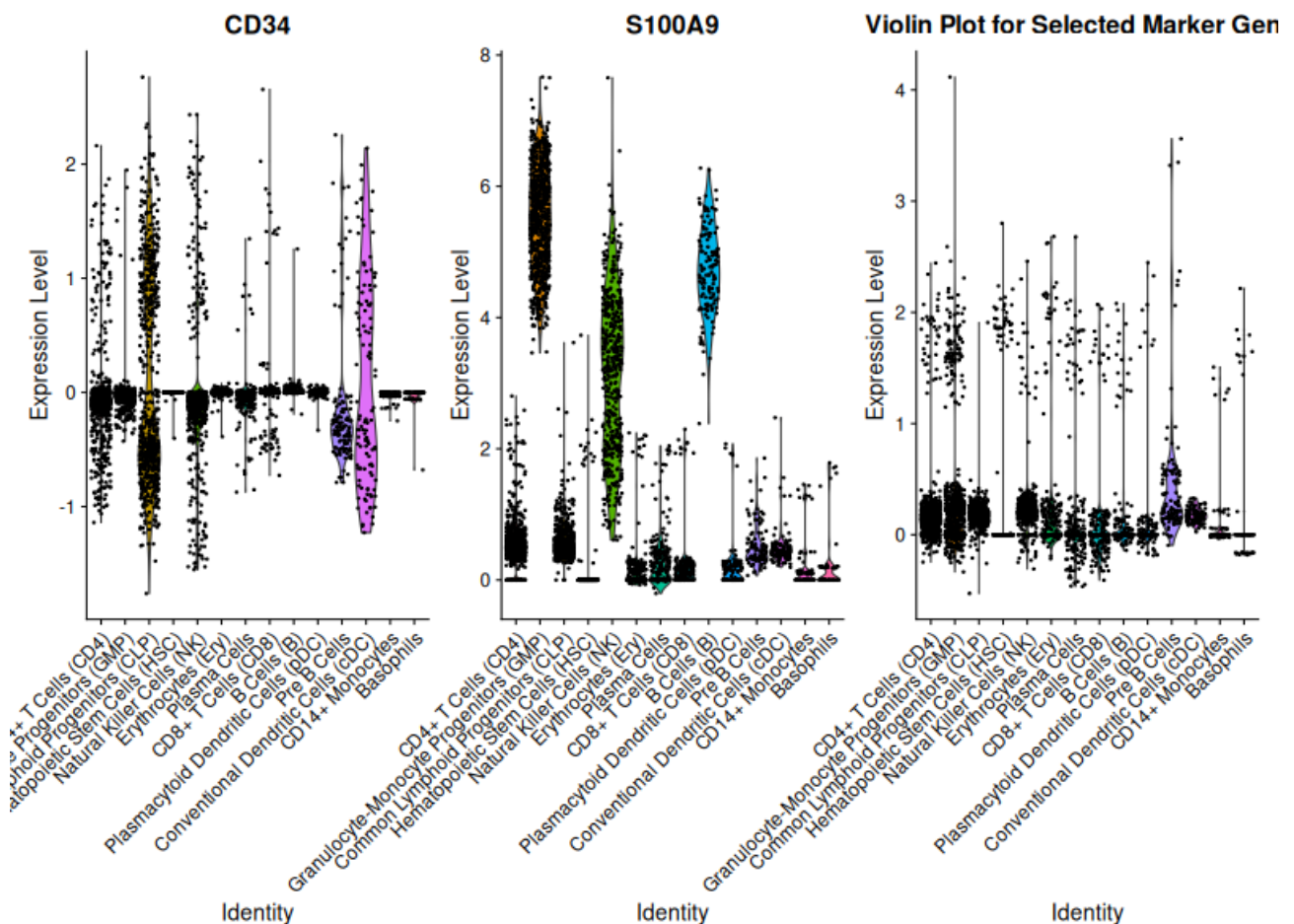
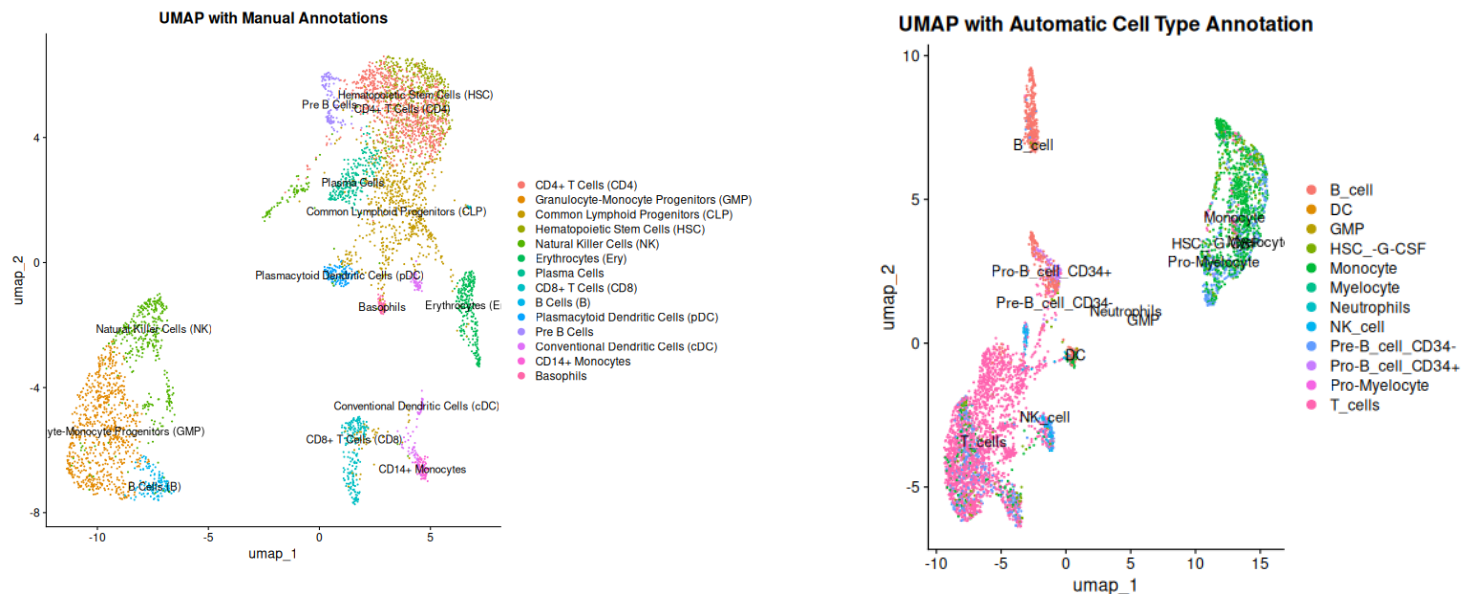


Manual annotaion



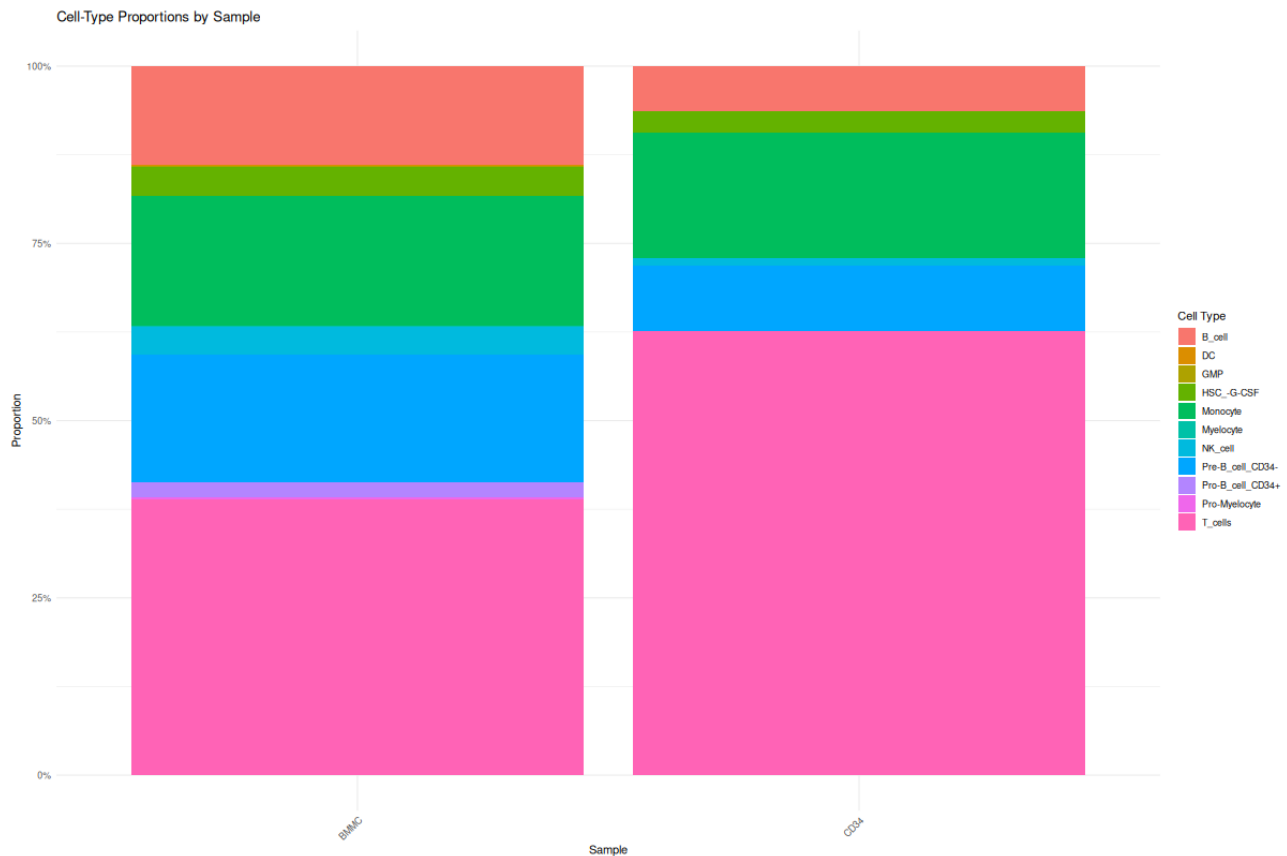
Comparison between manual & automatic annotation:

Form both figures we can see that there is slightly difference between automatic and manual annotation in number of clusters (there is more clusters in manual annotation)



-Gene expression :

Cell-Type Proportions



The standard deviation (variability) of proportions across samples demonstrates which cell types differ the most significantly:

T cells: The highest variability (0.168), indicating significant changes across samples.

Pre-B_cell_CD34- has moderate variability (0.062).

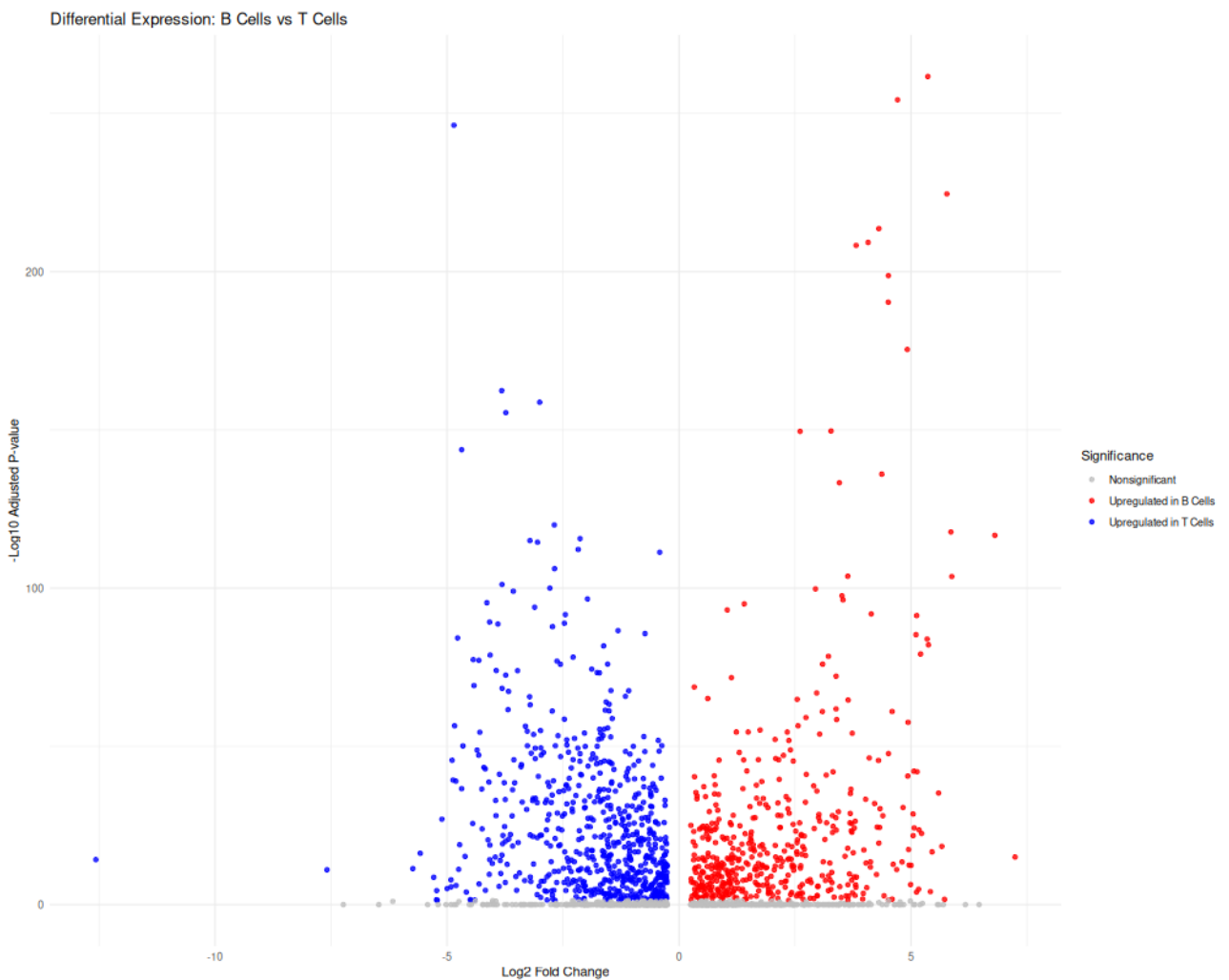
B cells exhibit lower variability (0.054).

NK cells and HSC_G-CSF exhibit minimal variability (0.020 and 0.008, respectively).

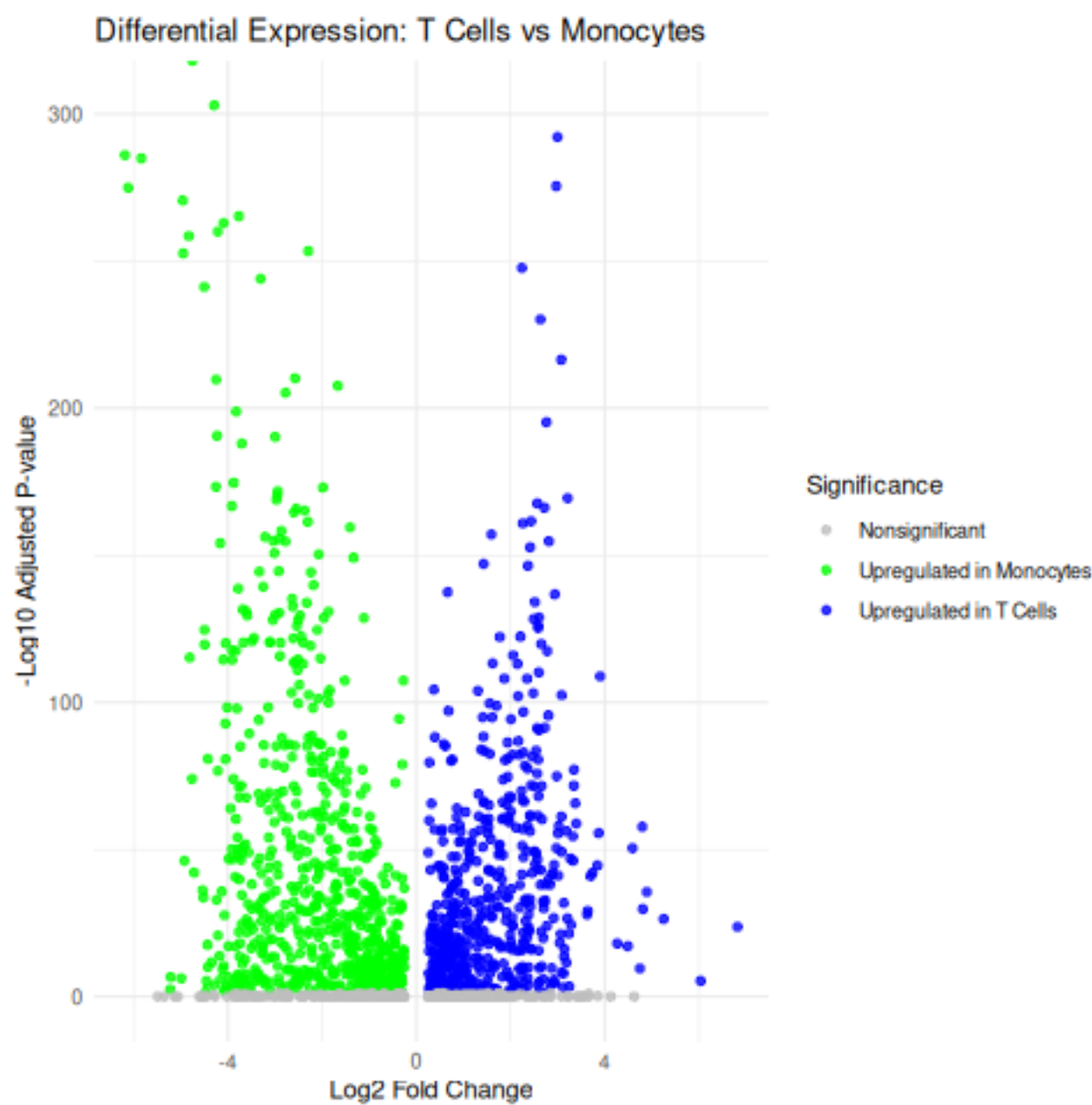
This shows that T cells account for the majority of variability between samples, whereas other cell types are more constant.

Differential Expression Analysis

1-B cells vs T cells :



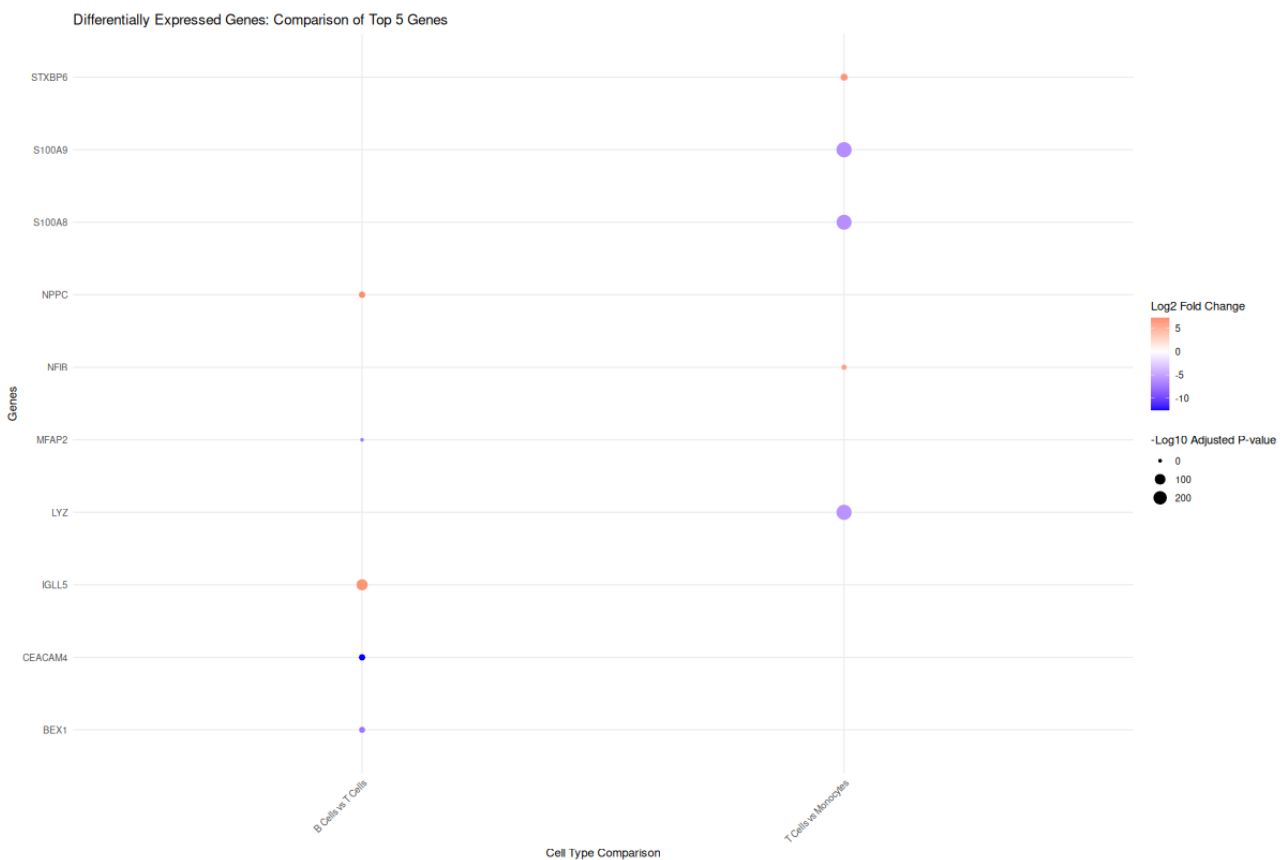
-T-cells vs Monocytes



- **Memory Formation on Cells :**

naive T cells and memory T cells provide distinct functions in the immune response. naive T cells are activated when exposed to new, unknown antigens. These cells trigger an immunological response, allowing the body to fight off infections for the first time. Memory T cells, on the other hand, originate from previously activated T cells and live in peripheral organs. They "remember" infections that the immune system has already encountered, letting it to respond quickly and robustly when exposed to the same antigen again.

- **Plot Differentially expressed genes**



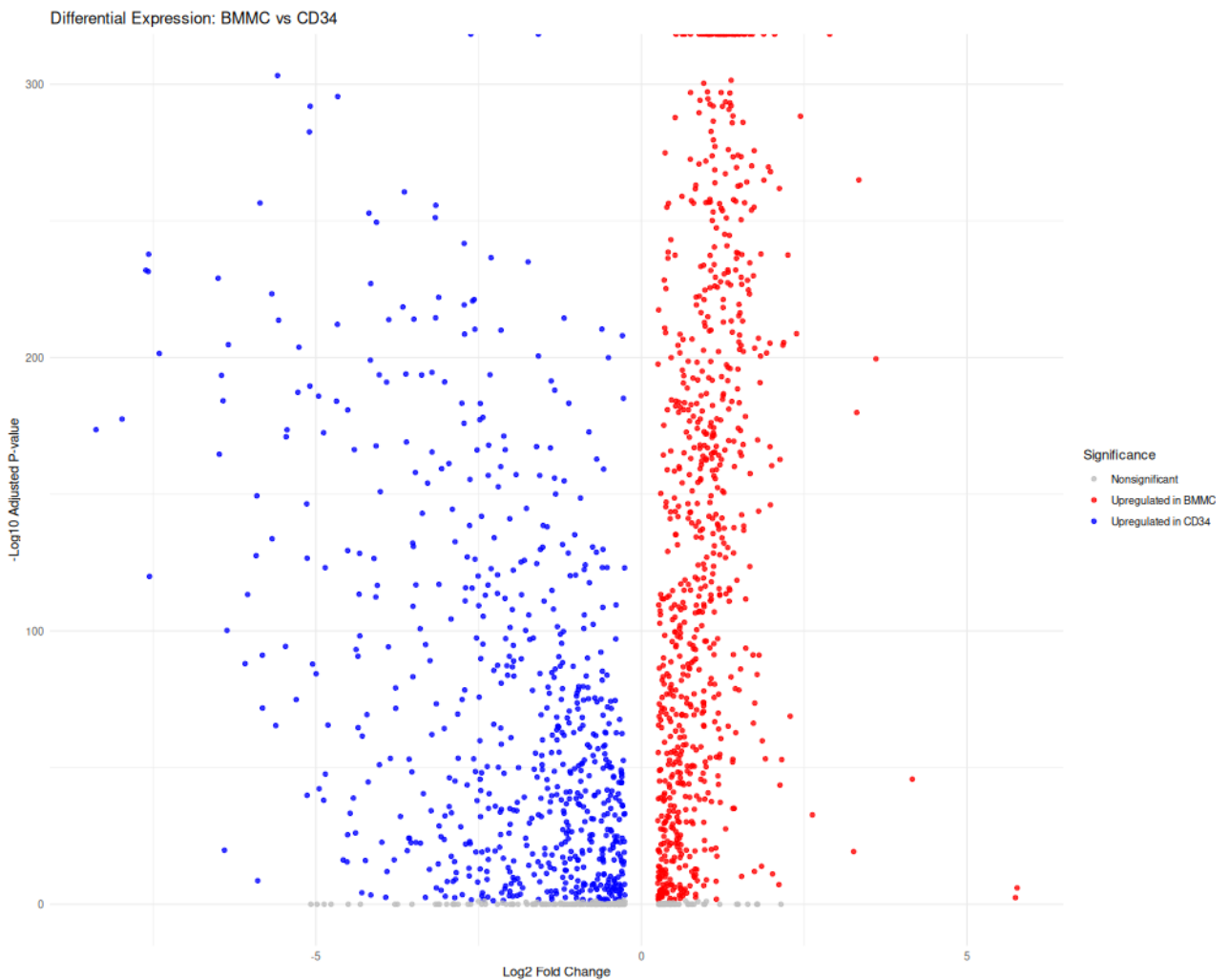
Pathway Analysis

8.1 Differential Expression Analysis on groups (2P)

the top 5 DEGs with p-value and Fold change

DE_results

GFI1B
RBPMS
SERPING1
CCDC175
ABCG2



Enhanced Volcano Plot

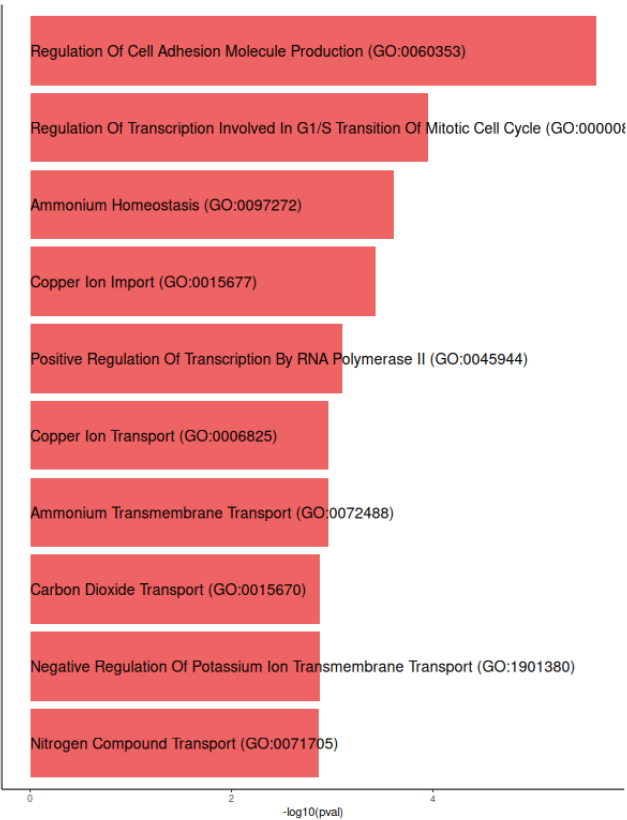


Pathway analysis on groups:

GO_Biological_Process_2023_BMMC_positive markers



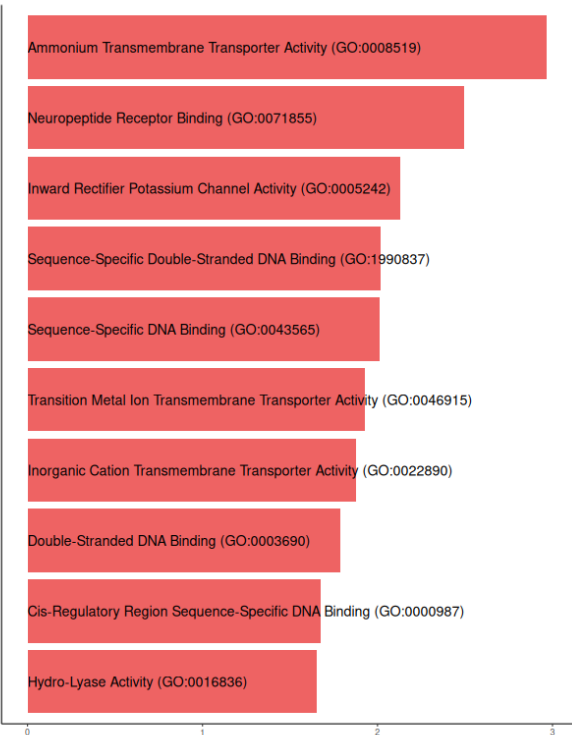
GO Biological Process: BMMC vs CD34



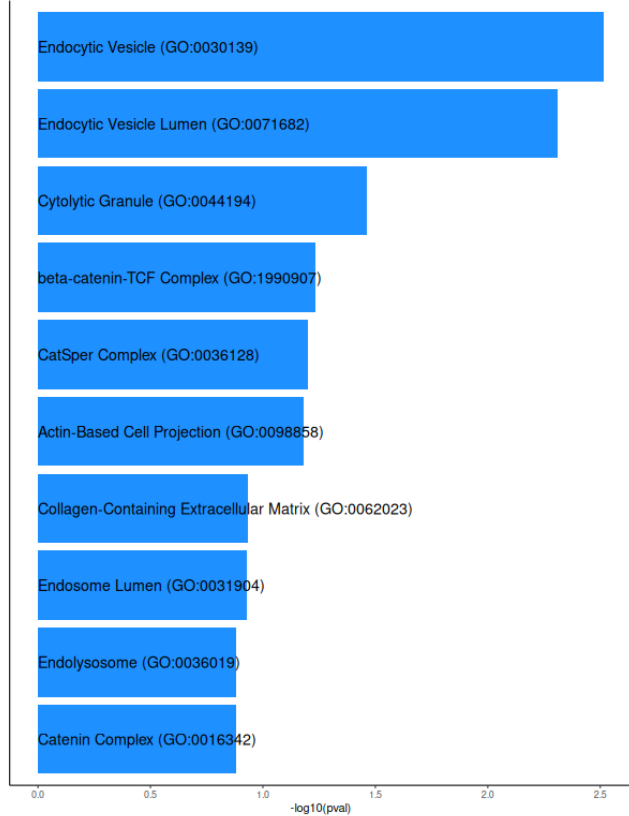
GO_Molecular_Function_2023_BMMC_positive markers



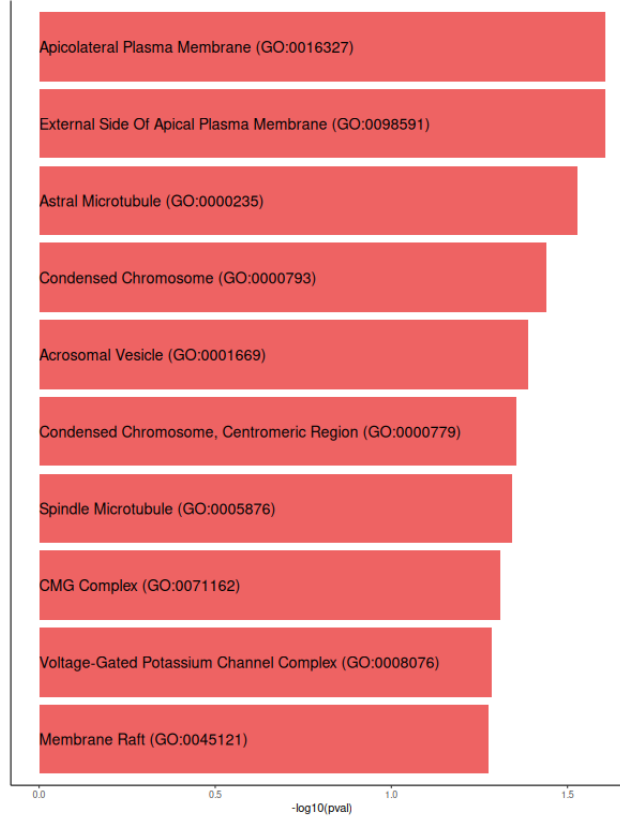
GO Molecular Function: BMMC vs CD34



GO_Cellular_Component_2023_BMMC_positive markers



GO Cellular Component: BMMC vs CD34



Biological interpretation:

The pathway with the lowest p-value is "mononuclear cell differentiation,"

which is statistically significant in the GO enrichment analysis. This biological process involves the development and differentiation of mononuclear cells,

which are critical components of the immune system, including lymphocytes and monocytes.

These cells play essential roles in immune response, inflammation, and tissue repair.

Top 20 Enriched GO Terms

